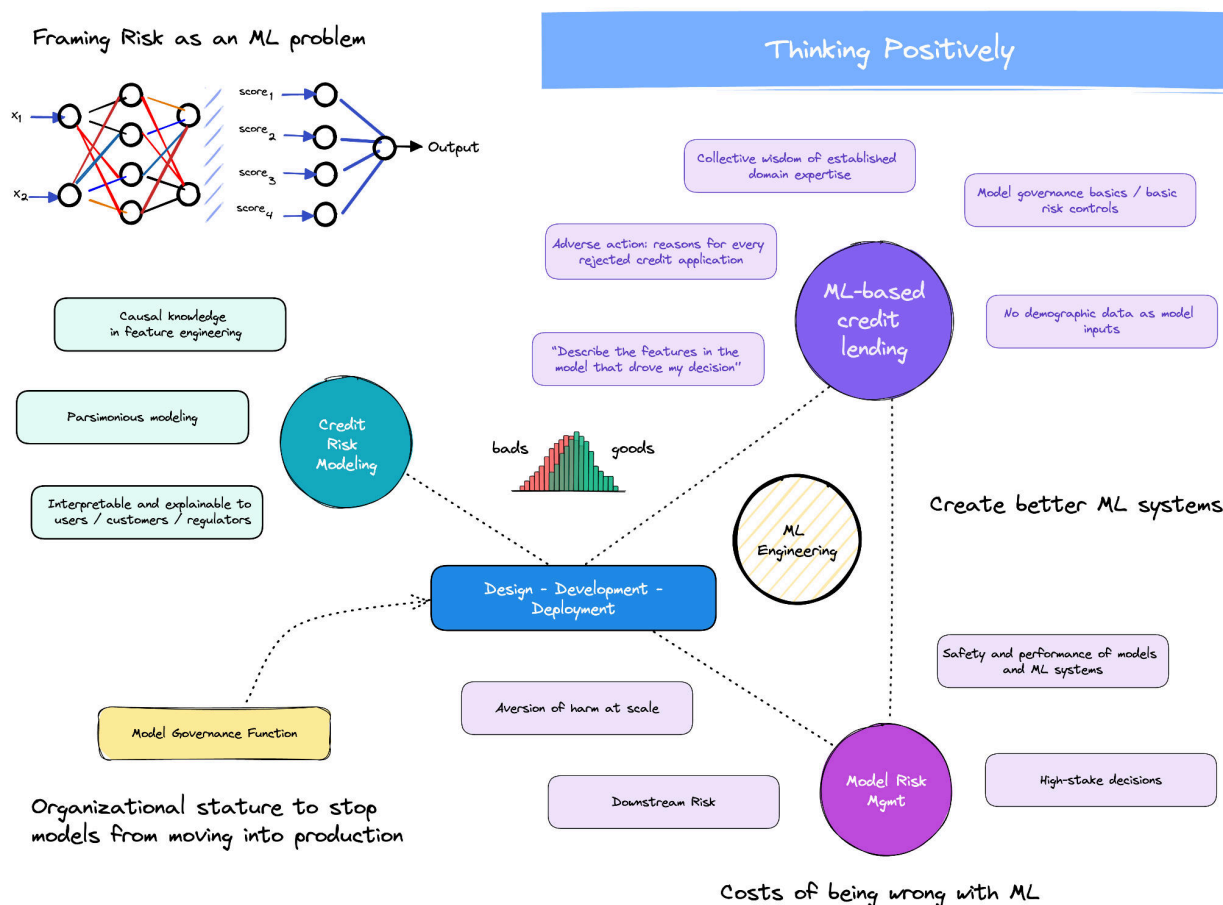


Lending, Credit Risk & Data Science

Notes collection compiled by Denis Burakov (<https://www.linkedin.com/in/denisburakov>)

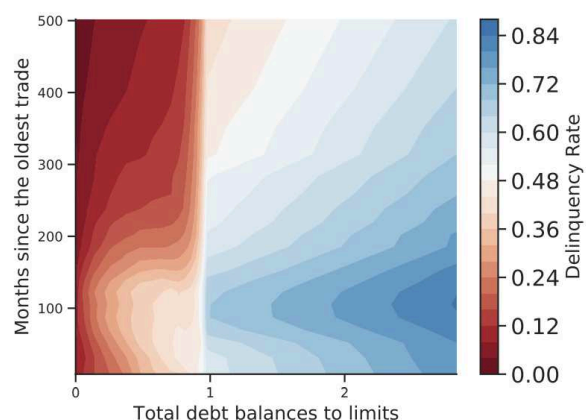
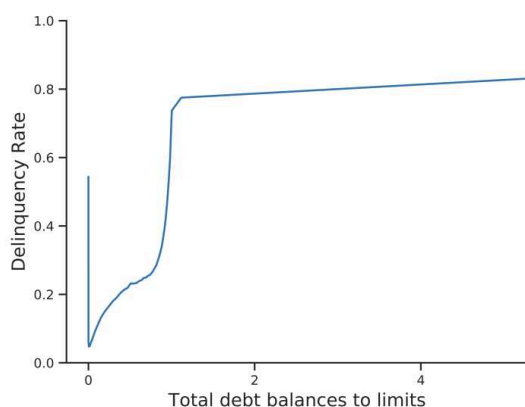
Credit Risk & Data Science

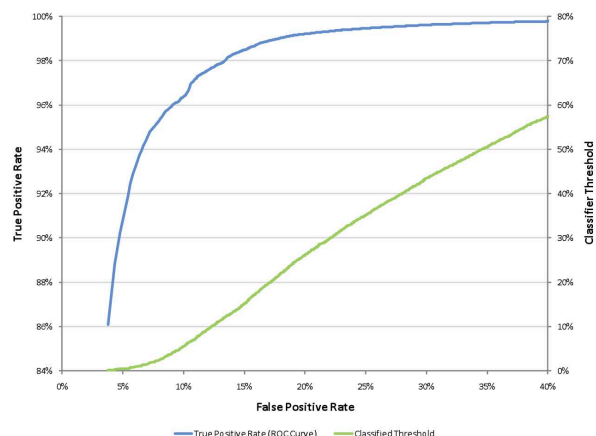
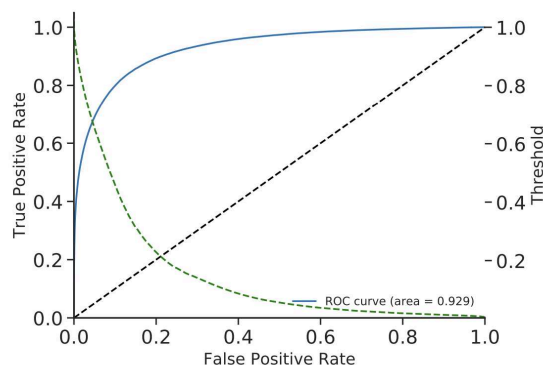
Credit Risk Management: Modeling, Strategy and Policy



- [Forecasting Loan Default in Europe with Machine Learning](#)
- [Credit Loan Decisions — Fairlearn 0.10.0.dev0 documentation](#)
- [Regulating AI Innovation: Aboitiz Data Innovation's David Hardoon](#)
- [Unlocking the power of model ops for risk management gains](#)
- [Adverse Selection on Maturity: Evidence from Online Consumer Credit](#)
- [Mortgage Credit Risk Transfer: Thriving Through The Cycle](#)

- [FICO Lessons in Developing, Applying Decision Modelling Methods - KDnuggets](#)
- [Towards Profitability: A Utility Approach to the Credit Scoring Problem](#)
- [Affordable Credit Poised to Save Consumers Billions | The Pew Charitable Trusts](#)
- [A Tutorial on the Economics of Payday Lending: Loan Sharking or Risky Lending](#)
- [Risk-Based Decisioning in an Age of Uncertainty Part 2 ↴](#)
 - Integrated single environment (historical or streaming data), data exploration simplifies data preparation enables to interact with data quickly (key relationship in the data)
 - In-house / external model development (same platform, takes the guess-work out of building a model) → drag and drop interfaces
 - Model validation, monitoring, model management, documentation → model performance KPIs, validation documentation
 - Intelligent decisioning with business logic (optimized strategies, data-driven outcomes must combine with policies) → real-time credit decision strategy
 - Data quality management, analytical modeling dataset creation, validation, back-testing, portfolio quality reporting on the same platform
 - The platform reduces Governance, implementation risks → brings unity and capabilities embedded
 - Centralized connected decisioning, essential for digital journeys
- [Wells Fargo Credit Modeling Model Card](#)
- [Consumer Credit Risk Models Via Machine-Learning Algorithms](#)
- [Predicting Consumer Default: A Deep Learning Approach | NBER ↴](#)



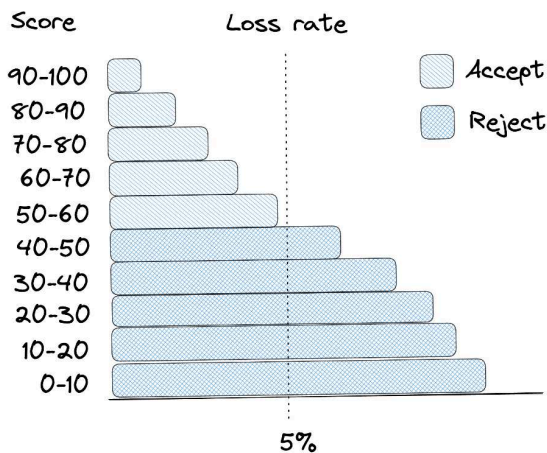


- [Model Comparison and Calibration Assessment User Guide for Consistent Scoring Functions in Machine Learning and Actuarial Practice](#)
- [ML & DS shades of Credit Risk Management. Part I. | by T1A](#)
- [Analytics In The Field Of Applied Credit Risk | by Nulearn | Medium](#)
- [Credit Risk Modeling with Machine Learning | by A. Jeremy Mahoney | Towards Data Science](#)
- [What Is Better: One General Model or Many Specialized Models?](#)
- [LGD Parameter Estimation for the Retail Mortgage Book Based on EBA Guidelines](#)
- [CONC 5.2A Creditworthiness assessment - FCA Handbook](#)
- [Data Visualization - Banking Case Study Example \(Part 2\) – YOU CANalytics](#)

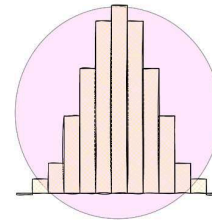
Credit scoring is a core competence of commercial banking. If competitors can exploit another bank's rating knowledge, they can improve their own profits at the expense of the bank whose knowledge has leaked. Therefore, banks should pay attention to whom they communicate their scoring methodologies.

Credit risk models must be robust to changing economic and regulatory environments.

Credit Policy Testing



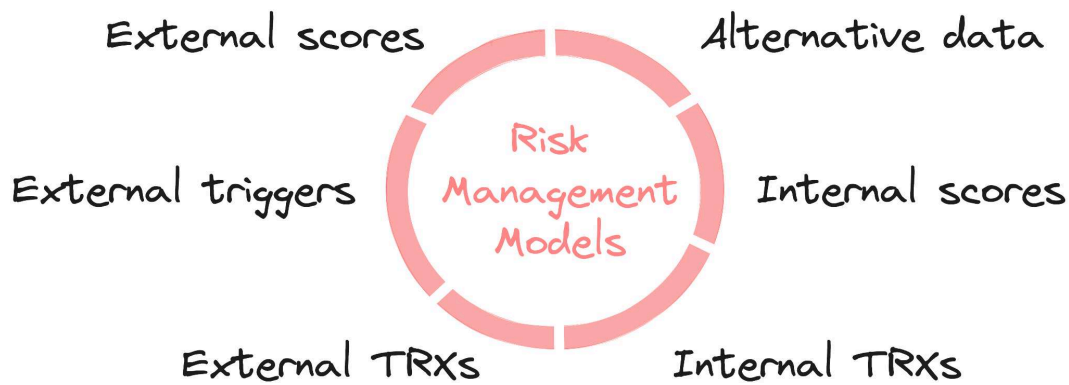
The loss rate must decrease monotonically with increasing scores



The number of quantiles increases as long as the loss rates are monotonic

ML Foundations in Credit Risk

- [Machine Learning in Retail Credit Risk: Algorithms, Infrastructure, and Alternative Data — Past, Present, and Future | NVIDIA On-Demand](#)
- [Machine learning in credit risk modeling : a James white paper](#)
- [Machine learning explainability in finance: an application to default risk analysis | Bank of England](#)
- [Explaining and Accelerating Machine Learning for Loan Delinquencies | NVIDIA Technical Blog](#)
- [Accelerating Trustworthy AI for Credit Risk Management | NVIDIA Technical Blog](#)
- [Credit Scoring Software | AI Credit Score Solution | GiniMachine](#)
- [Awesome-credit-modeling](#)
- [Machine Learning approach for Credit Scoring](#)
- [Automating Interpretable Machine Learning Scorecards \[Moody's Analytics\]](#)
- [The Use of Machine Learning for Credit Underwriting](#)



Net Present Value in Lending

There is a "sweet spot" on the risk gradient that has large enough credit limits, long enough loan lives, and enough revolvers to accumulate finance charges that the borrowers subsequently pay back.

I think it was Scoreplus who introduced a Cleverness Index, calculated as the average profit on a Good account divided by the average loss on a Bad account. Working to improve the value of this index, either across the board or in segments, is the key to improving profitability; and loan or limit management is a key tool in this.

- [US7383215B1 - Data center for account management - Google Patents](#)
- [Credit Card Market Study: Interim Report: Annex 5 – Firm business model analysis](#)
- [How To Assign FTP Attribution to a Loan](#)
- [Loss rate forecasting framework based on macroeconomic changes](#)
- [What Charge-Off Rates Are Predictable by Macroeconomic Latent ...](#)
- [Profit scoring for credit unions using the multilayer perceptron, XGBoost and TabNet algorithms: Evidence from Peru - ScienceDirect](#)
- [Upstart Macro Index - UMI](#)
- [Parallel Timing Curve Calibration ↴](#)

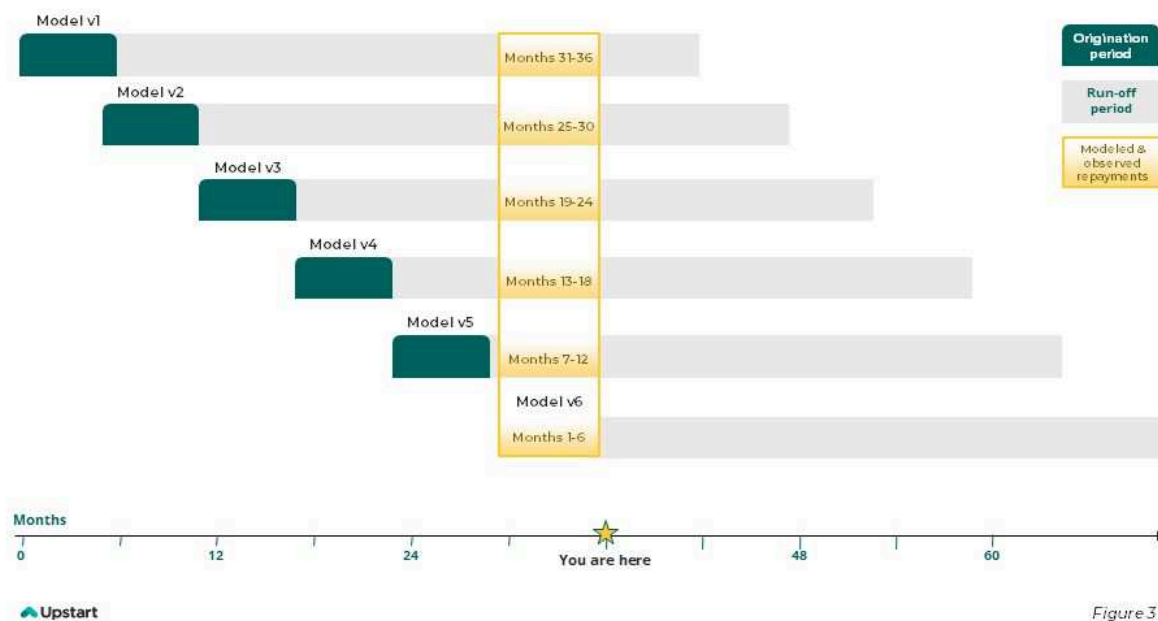
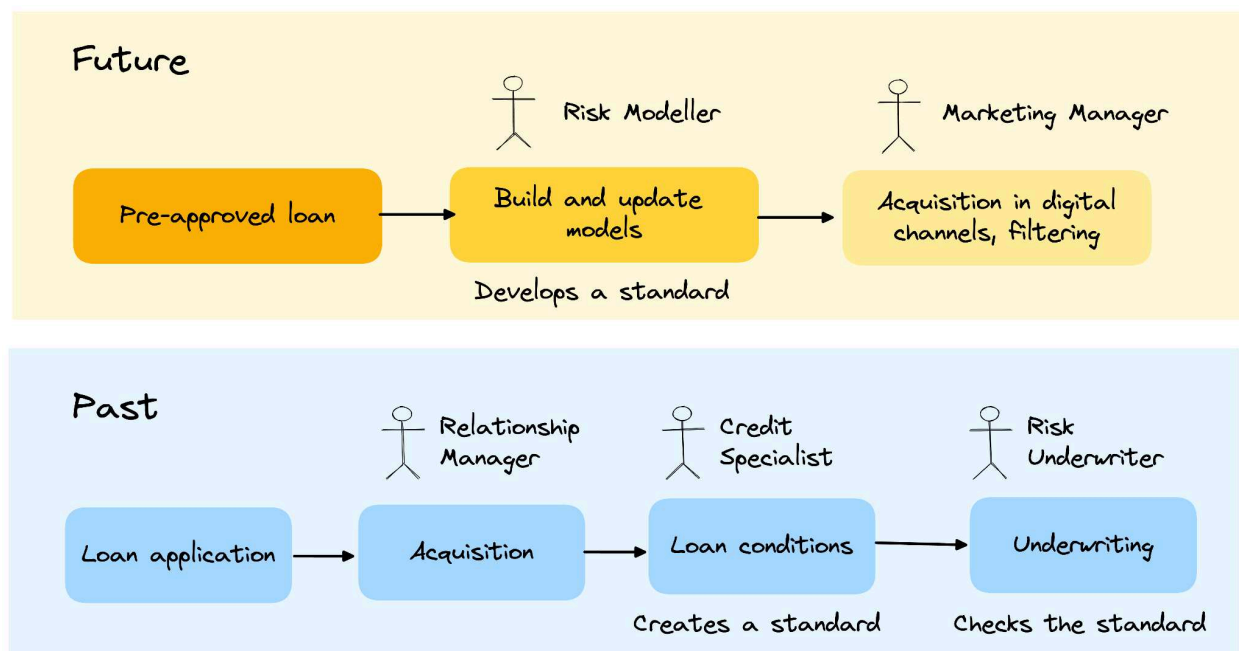


Figure 3

First, we'll go back in time to the launch of Model v6. While preparing to launch our new model, we ask the model to evaluate all of the outstanding loans on the platform and predict the likelihood of default or prepayment for the remaining months on each of those loans. We call this "re-underwriting." It is not a backtest. The predictions made across the platform are for payment months that have yet to occur.

As a result, at month six we can observe how accurately the model predicted repayments across the entire timing curve rather than just the first six months (Figure 3). Our feedback loop has shrunk from three years to, in theory, just one month. But, in practice, it may require a few months to ensure that the single month isn't an outlier.



Profit Scoring

There is a "sweet spot" on the risk gradient that has large enough credit limits, long enough loan lives, and enough revolvers to accumulate finance charges that the borrowers subsequently pay back.

While a high-end portfolio consisting mainly of transactors may have a close correspondence between profit and risk, this is not the case in lower-end revolver segments where the increase in risk is disproportionate to the increase in profit when switching to profit scoring. These results reinforce the well-known importance of risk-based guardrails when using new modeling methodologies in underwriting for issuers concentrating on higher-risk customers.

- [Machine Learning with SAP HANA & R – Evaluate the Business Value](#)
- [User Churn Prediction: A Machine Learning Example](#)
- [Four tips for predicting and preventing churn | by BearingPoint | Medium](#)
- [A Modified AUC for Training Convolutional Neural Networks: Taking Confidence Into Account - PMC](#)
- [Explainable Machine Learning in Credit Risk Management](#)
- [On the risk-adjusted performance of machine learning models in credit default prediction](#)

- [Profitability vs. Credit Score Models—A New Approach to Short Term Credit in the UK](#)
-  [Bondora Peer to Peer Lending Loan Data | Kaggle](#)
- [Bondora Peer-to-Peer Lending Data | IEEE DataPort](#)
- [Public Reports | Bondora](#)
- [General statistics | Bondora](#)
- [Economic benefit of powerful credit scoring](#)
- [The use of profit scoring as an alternative to credit scoring systems in peer-to-peer \(P2P\) lending](#)
- [Application scoring dataset from Cost-sensitive learning for profit-driven credit-scoring](#)
- [Full article: Cost-sensitive learning for profit-driven credit scoring](#)
-  [A boosted decision tree approach using Bayesian hyper-parameter optimization for credit scoring](#)
- [Moving beyond alternative data — cash flow underwriting becomes industry standard | Plaid](#)
- [Cost-sensitive Learning for Profit-driven Credit Scoring](#)

Inventory Finance

- [Inventory financing under Risk-Adjusted-Return-On-Capital criterion](#)
- [How to use inventory financing to improve your cash flow - Ramp](#)
- [How Ramp Accelerated Machine Learning Development to Simplify Finance](#)

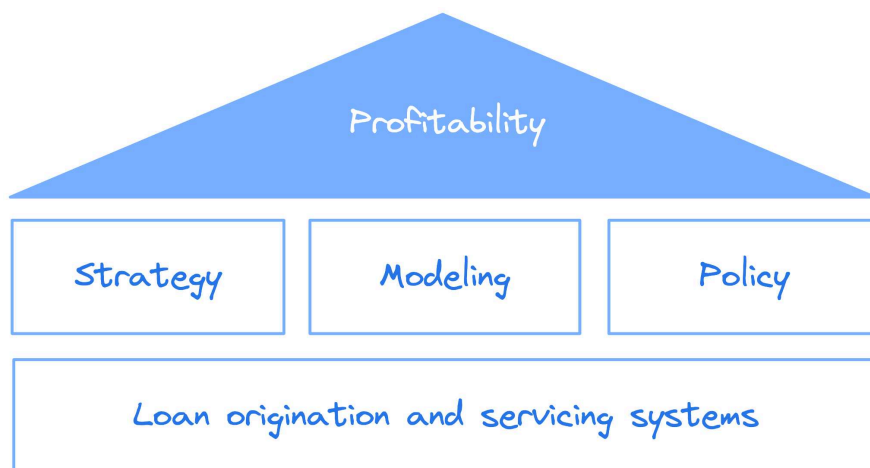
P2P Lending

- Most of lenders have little knowledge on credit risk management and are unable to build a portfolio to maximize the expected value of investment return
- Unprofessional lenders usually fail to build a rational portfolio
- Most lenders lack financial expertise and professional skills to screen potential high-risk borrowers

[We are in an early adoption phase of AI/ML technology in the provision of financial services](#)

Institutions prioritize the ease of interpretability and explainability of results over the potential gains on predictive power (managing the tech risk of sufficient performance vs interpretability and user value how fast we can ship the product + market fit)

Credit risk is the most significant risk by impact for any bank and financial institution. Accurate credit risk assessment affects an organization's balance sheet and income statement, since credit risk strategy determines pricing, and might even influence seemingly unrelated domains, e.g. marketing, and decision-making.



Precise credit risk estimation is key to an organization's profitability. If the bank fails to estimate a risk correctly, it either overprices loans and loses its market share, or sets interest rates too low to cover the expected losses, which leads to poor financial results.

Lending & FinTech

The Kickstarter problem is the gap that exists, in the age of the internet, between prospective customers' enthusiasm for a product concept and the ability of an entrepreneur to actually deliver a fully functional version of that product at scale.

The loan servicing systems govern that post-origination experience. Loan servicing is the process of administering the full, post-origination lifespan of a loan, from the day the account is created to the day that it is closed and the balance is fully repaid.

The best loan servicing systems are built on intellectual property that helps companies understand how to ensure their lending product concepts are compliant.

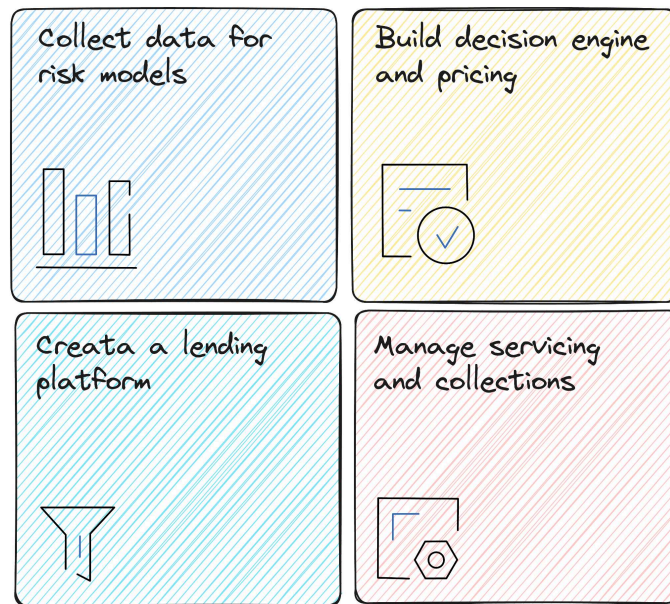
[Embedded Finance: What It Takes to Prosper in the New Value Chain | Bain & Company](#)

- Create a better borrower experience
- Foster a more interconnected ecosystem
- Build adaptable lending products

Loans are enablers: they don't have a purpose in and of themselves. They only exist to get consumers and businesses what they want (a house, a new oven for your bakery, etc.) As such, anything that financial services providers can do to reduce the effort that a customer has to go through to get a loan will be immensely valuable to that customer because it shortens the time until that customer gets their hands on the thing they actually want.

- [Unlocking the Next Wave of Lending Innovation](#)
- [5 Crucial Steps to Building a Lending Product](#)
- [Reinventing credit cards: Responses to new lending models in the US | McKinsey](#)
- [Upgrade Card Review 2023: Cash Rewards Visa – Forbes Advisor](#)
- [Guide to the Upgrade Card | CreditCards.com](#)
- [New credit-risk models for the unbanked | McKinsey](#)
- [📺 How to Build a Profitable Digital Bank with Tinkoff](#)
- [Fodder for Future Neobank Founders - by Samora Kariuki](#)
- [The Opportunity with AI Driven Lending: Paul Gu](#)
- [Safe Is Risky: How Banks Can Stay Innovative](#)
- [A.I. has a discrimination problem. In banking, the consequences can be severe](#)
- [10 Themes For European Retail Banking 2023](#)
- [How Fintech Startups Can Pivot To Profitability Despite Any Recession](#)
- [Consumer lending strategies FIs can use to drive growth | EY - US](#)
- [How banks can reimagine lending to small and medium-size enterprises | McKinsey](#)
- [Overpromised And Underdelivered: Is The Fintech Wave Still Coming?](#)
- [Financing Inventory Purchases Using Kickfurther Crowdfunding Platform](#)
- [What Is Inventory Financing? How Does It Work? | Lantern by SoFi](#)
- [Understanding Nubank - by Tanay Jaipuria](#)
- [F-1 Nubank US](#)
- [The LT3000 Blog: The capacity to suffer, LCV/CAC, and Capital One Financial](#)

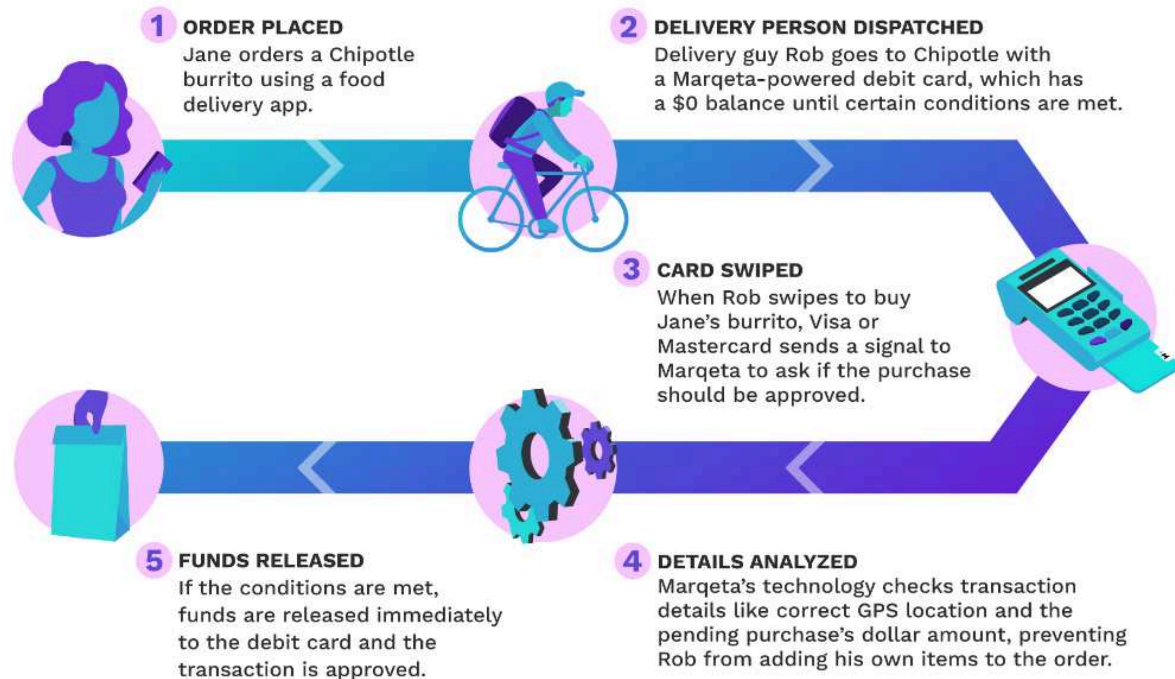
Operations to manage Credit Risk



Loan default rates can significantly vary across different portfolios and markets, which highlights the importance of effective strategies for handling data imbalance in credit scoring. One commonly used technique is resampling. Resampling involves either undersampling the majority class (non-defaulters) or upsampling the minority class (defaulters) to create a balanced label distribution for model training.

However, this strategy has two primary drawbacks: potential loss of vital information from the majority class and calibration issues, leading to meaningless interpretation of predicted PDs without recalibration. Regardless of the chosen method (upsampling the minority or downsampling the majority), it's crucial to note that no strategy can magically resolve the problem of having low absolute numbers of defaults and other approaches like tuning the cut-off threshold based on a business metric can be more preferable.

Marqeta's Money Moves



NICK DESANTIS

To make money on a product we try to minimize output and maximize outcomes.

We use product discovery work to identify real customer needs and really test that we'll get the outcome we hope for.

Deep collaboration with engineering and design in order to combine enabling technology with exceptional user experience in ways that provide real value to the customers and to your business.

The product manager is responsible for value and viability, the designer is responsible for usability and the engineers are responsible for feasibility. Together we make a product.

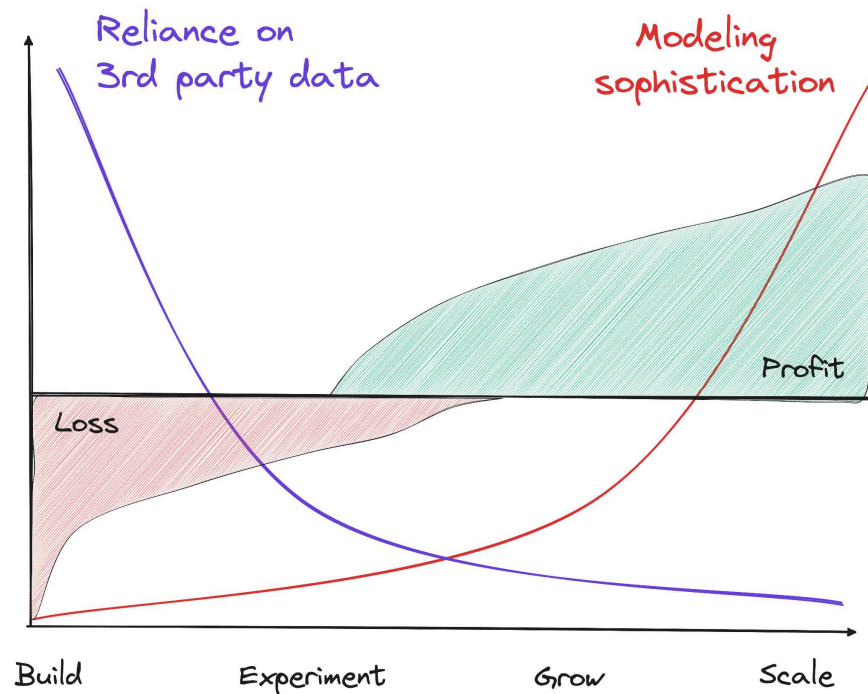
Seamless delivery of financial services

- To help people and businesses to invest money to provide for their financial needs in an uncertain future
- Part of the appeal of machine learning is that it can make the very time-consuming activity of risk management more efficient or cheaper

- At every stage of the machine learning process, errors can undermine the accuracy of risk assessments
- Implementing this kind of safeguard may become increasingly difficult, though, as risk modeling technology moves into areas of data science and computer programming that are outside of the traditional expertise of financial institution employees
- A seminal study focused on a machine learning model that had learned that any patient with asthma had a reduced risk of dying from pneumonia. This in fact turned out to be true, but only because asthma was such a dangerous precondition for pneumonia that patients with asthma were treated earlier and better—not because having asthma itself makes pneumonia less harmful
- Magnify the impact of risk management errors
- Automated financial future

[It Took Three Tries, But This Entrepreneur Transformed His Struggling Startup Into A \\$2 Billion Unicorn](#)

- [A lenders' guide to achieving positive unit economics](#)
- [Unit Economics of Lending & how Tech can improve them \(Part 2/3\)](#)
- [How we think about Risk at Capchase](#)
- [Building a Lending Business from Scratch](#)
- [Credolab takes unique approach to credit scoring, financial inclusion | News](#)
- [The Capital One Data Science Interview | by Terence Shin](#)



Data, Product & Engineering

- [To Really 'Disrupt,' Tech Needs to Listen to Actual Researchers | WIRED](#)
- [APIs Explained for Product Managers](#)
- [What are the risks and disadvantages of Data Mesh? | Xomnia](#)
- [The Rise of the AI Engineer - Latent Space](#)
- [Modernizing our Data Platform - DataRunner](#)
- [Conway's law](#)
- [5 Founders on the Future of Data | Andreessen Horowitz](#)
- [The modern data stack: Tim Harrison speaks to Tristan Handy](#)
- [Catalog & Cocktails: What does a Chief Data Officer do? on Apple Podcasts](#)
- [Are You Asking Too Much of Your Chief Data Officer?](#)
- [Why Do Chief Data Officers Have Such Short Tenures?](#)



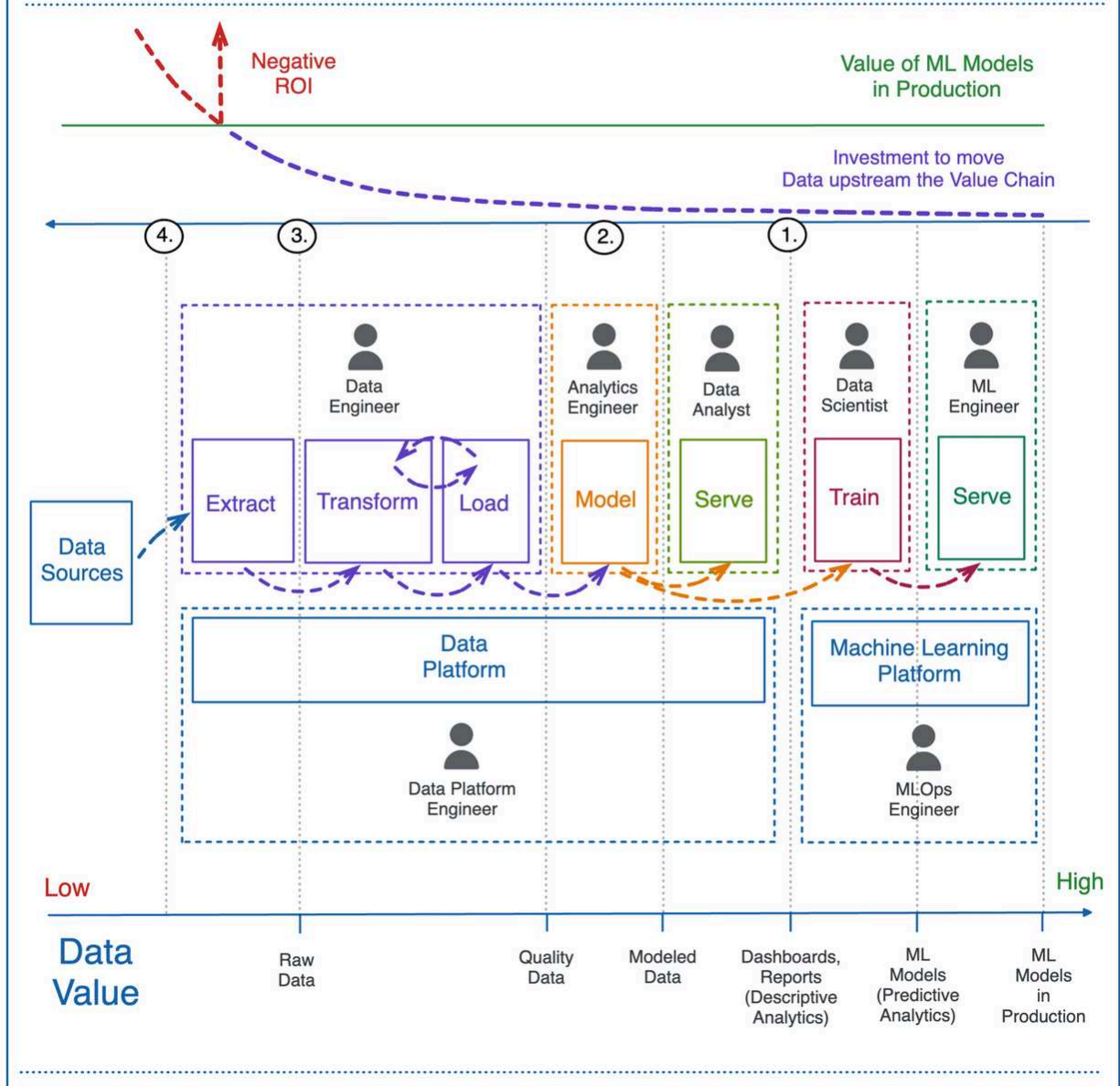
linkedin.com/in/aurimas-griciunas

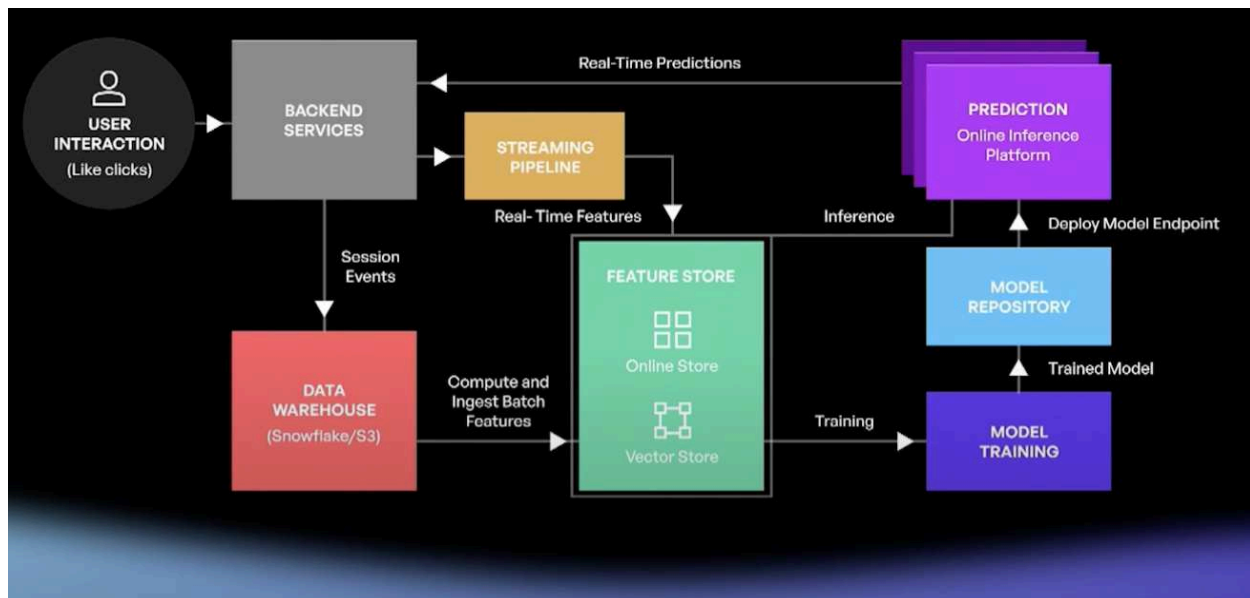
@Aurimas_Gr

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www.swirlai.com

The Data Value Chain





Hyper-Scaling Real-Time Personalization with Privacy via On-Device Computing

Data Teams

We need to meet these expectations with an approach and architecture that natively supports diverse use cases and does not require copying data from one technology stack to another across the organization so that we can meet the needs of yet another use case.

Operational data sits in databases of microservices, applications, or systems of records that support the business capabilities.

- Operational data is used to run the business and serve the users
- Analytical data is used to optimize the business and user experience

This is the data that fuels the AI and analytics aspirations.

The future of intelligent banking requires data beyond the financial transactions that customers perform with their banks. They'll need to know the customers' housing needs, the housing market, their shopping habits, their dreams, to offer them the services they need, when they need it.

Domain ownership distribution results in a distributed data architecture, where the data artifacts - datasets, code, metadata, and data policies - are maintained by their corresponding domains.

Removing centralized bottlenecks

A centralized data team, managing a monolithic data lake or warehouse limits agility, particularly as the number of sources to onboard or number of use cases grow. Data Mesh

looks carefully for centralized bottlenecks, particularly where they are the focal point of multi-party synchronization, both from the human communication perspective and architecture. These bottlenecks include data lakes and data warehouses.

They took the microservice architecture and the DevOps way of working and applied it to the data realm creating the data mesh. The data mesh can be thought of in a simplistic way as a microservice architecture where a microservice is a data warehouse.

Data Engineer: inquisitive nature & mind / test & learn / translating Tech lingo into business opportunities

Data Artists instead of Data Scientists?

Focus: revenue, operational efficiency, risk management

- Regulation is the outcome
- Product teams: adoption, Services teams: focus on shipping the solution (don't care if it's used or not)

 [The Data Science Career Ladder](#)

Modern Data Stack

Data Analytics is about building knowledge about the organization. Data engineering will evolve to become business engineering by 2030 turning into a new discipline.

Operational characteristics being low-level: "operating forklifts" e.g. parquet file in this S3 bucket → not a good use of anyone's time. SQL = database language that acts like a calculus (Edgar Frank "Ted" Codd at IBM 1970s). It's not complete for relational semantics, table is not the best representation for data.

Dynamic nature of compute load (classic enterprise resource requirements are predicted, operational constraints too → you know how costly it is to run).

Modern data stack → data was scattered everywhere, DE/DS job was to wrangle and source data, MDS puts everything in one place, including access controls. Now we can do higher-order concepts, we needed dbt in the data processing stack (part of MDS).

- [The arc of data innovation with Bob Muglia](#)
- [The Datapreneurs](#)
- [Data engineers + dbt v1.5: Evolving the craft for scale | dbt Developer Blog](#)
- [Leveraging dbt and Python for Predictive Modeling | by Denny | Jul, 2023 | Medium](#)
-  [The modern data stack sucks](#)

Data Mesh

The State of Data Mesh (Notes from Zhamak Dehghani "Data Mesh" and interviews)

Data Mesh Principles and Logical Architecture

Storage | Compute | *Future data developer experience*

Domain Teams: Consumers and Producers at the same time

Org structure pulls teams apart → more conversations about org structures, Mesh economics (study of incentives)

Zhamak: Find lead adopters (feeling the pain most, start there and show value there), *don't build data products within centralized Data Teams → they will stay there, they will not leave*

"Data Mesh actually is not a thing, it's just a decentralized way of controlling data, decentralized ownership, responsible sharing of data (fairer future for people who want to participate)"

Chapter 5: organizational topics of Data Mesh adoption

Central Team

Mentorship and facilitator of a data product being built, something akin to a centralized committee. Interoperability → key focus of these efforts

Central team is useful in managing smaller domains with very limited data (find how we can align how business works), crucial to upping the capabilities

Rearchitecting teams (backwards Conway's Law), Siloed world vs Data Mesh, domain boundaries

Increasing motivation | decreasing / lowering friction

Data Office at Lego:

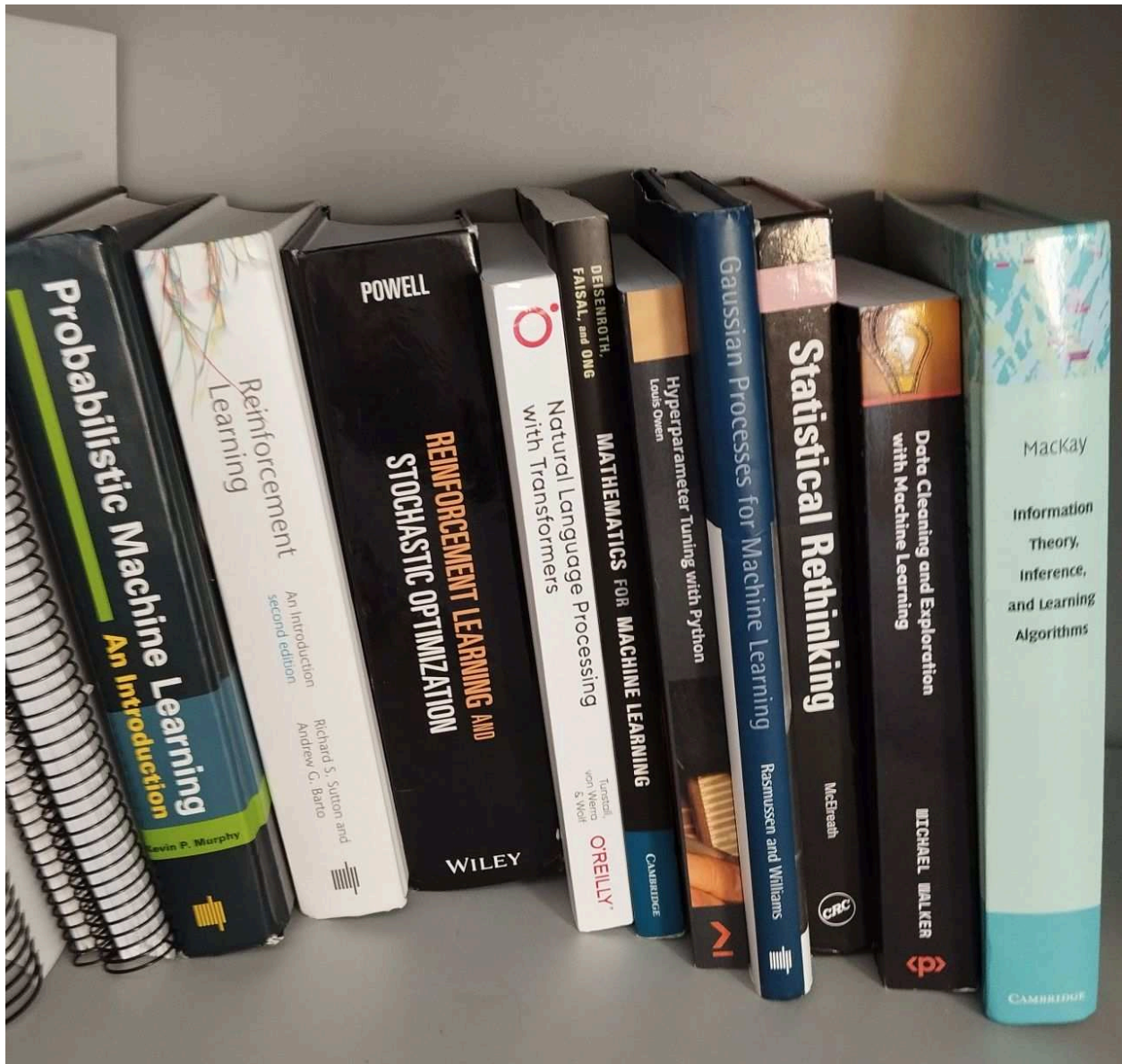
Data for enablement, data to innovate faster. Anonymize data → get started right away (same output)

AI Product Management

- [Becoming a Data Product Manager – DataTalks.Club](#)
- [Why Your Company Needs Data-Product Managers](#)
- [Designing Great ML Experiences - WWDC19 - Videos - Apple Developer](#)
- [Building machine learning products: a problem well-defined is a problem half-solved.](#)
- [Understanding the Data Landscape and Strategic Play Through Wardley Mapping](#)

- 
- [Simulating Intelligence - Library - Google Design](#)
 - [Unlocking the Business Value of Machine Learning—With Organizational Learning | AWS Cloud Enterprise Strategy Blog](#)
 - [How to Stop Delegating and Start Teaching](#)
 - [APIs Explained for Product Managers](#)
 - [Why 'burning bridges' may not be a career killer - BBC Worklife](#)
 - [Building AI that works: How to navigate the AI frontier as a product leader - The Intercom Blog](#)
 - [Slow Down to Speed Up Your Product Analytics](#)

Machine Learning & Data Science

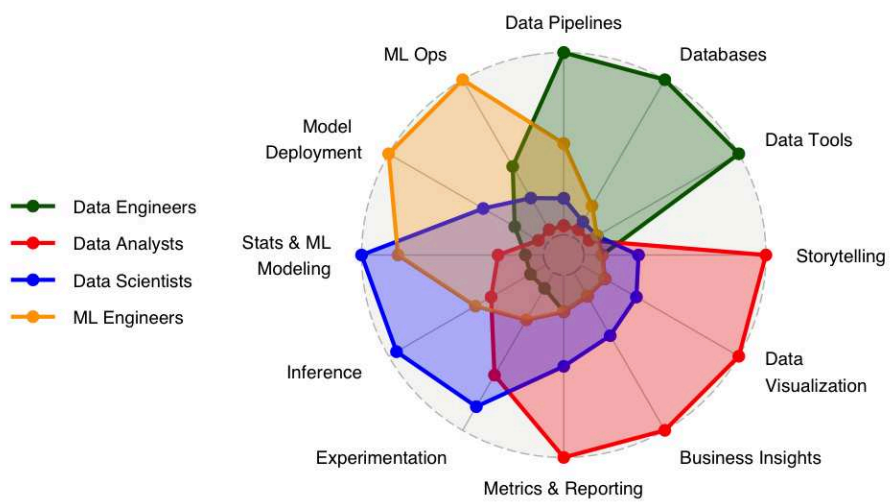
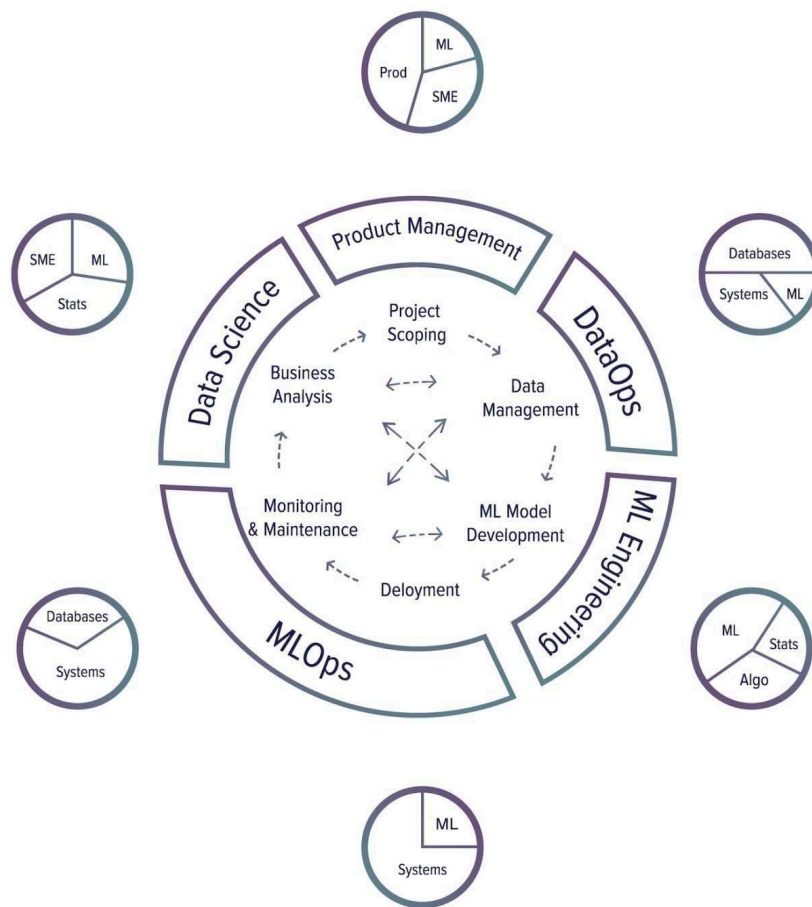


- [Machine Learning Engineer Learning Path | Google Cloud Skills Boost](#)
- [Practical Lessons from Predicting Clicks on Ads at Facebook](#)
- [How Facebook encodes your videos - Engineering at Meta](#)
- [\(Stochastic\) Gradient Descent, Gradient Boosting — Applied Machine Learning in Python](#)
- [aml-09-gradient-boosting-calibration](#)

- 
- [Understanding Random Forests: From Theory to Practice](#)
 - [Calibration of probabilities for tree-based models | Guilherme's Blog](#)
 - [Beyond Discrimination: a Comparison of Calibration Methods and Clinical Usefulness of Predictive Models of Readmission Risk - PMC](#)
 - [Classifier calibration: a survey on how to assess and improve predicted class probabilities | Machine Learning](#)
 - [A Unified View of Performance Metrics: Translating Threshold Choice into Expected Classification Loss](#)
 - [Forecasting with Economic News](#)
 - [Automated Linking of Historical Data | NBER](#)
 - [Which supervised learning algorithms are available for matching? - Data Science Stack Exchange](#)
 - [Explaining Predictive Model Performance: An Experimental Study of Data Preparation and Model Choice](#)

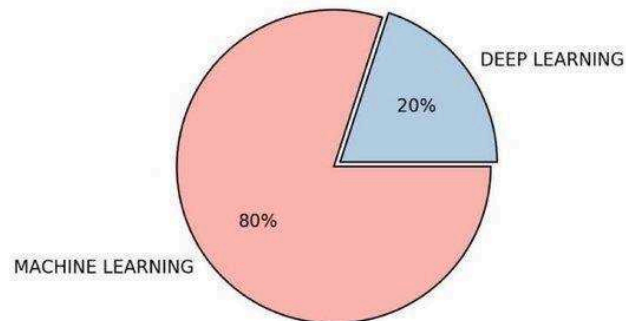
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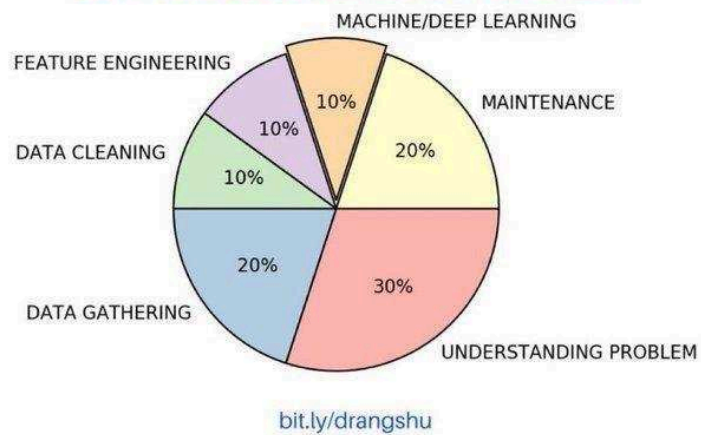
DATA SCIENTIST JOB - EXPECTATION

@drangshu

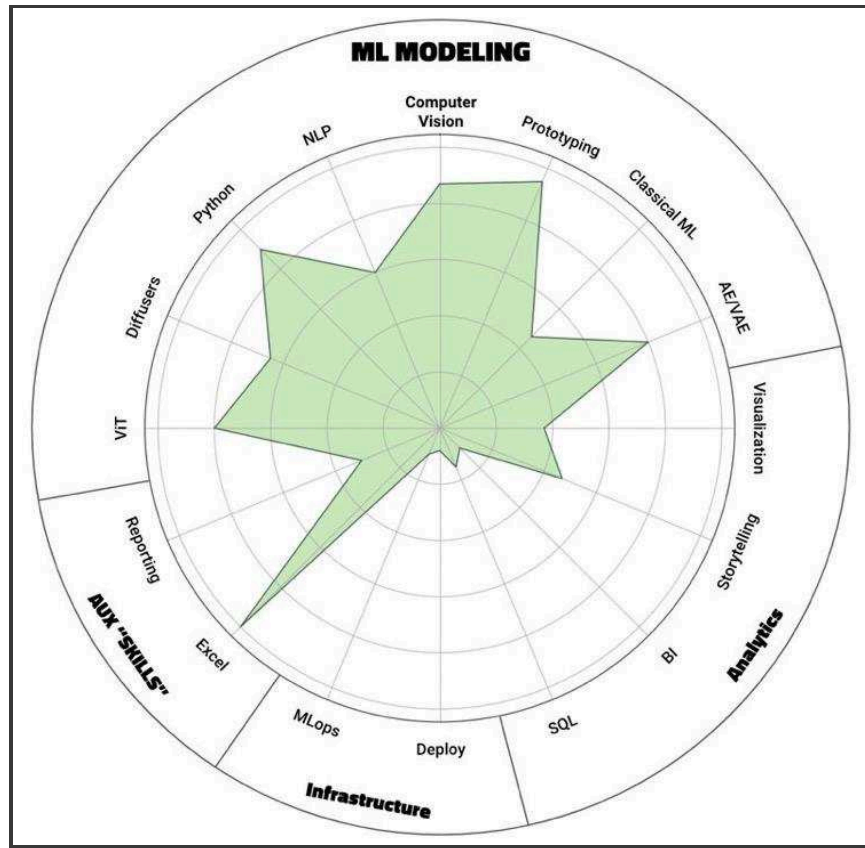


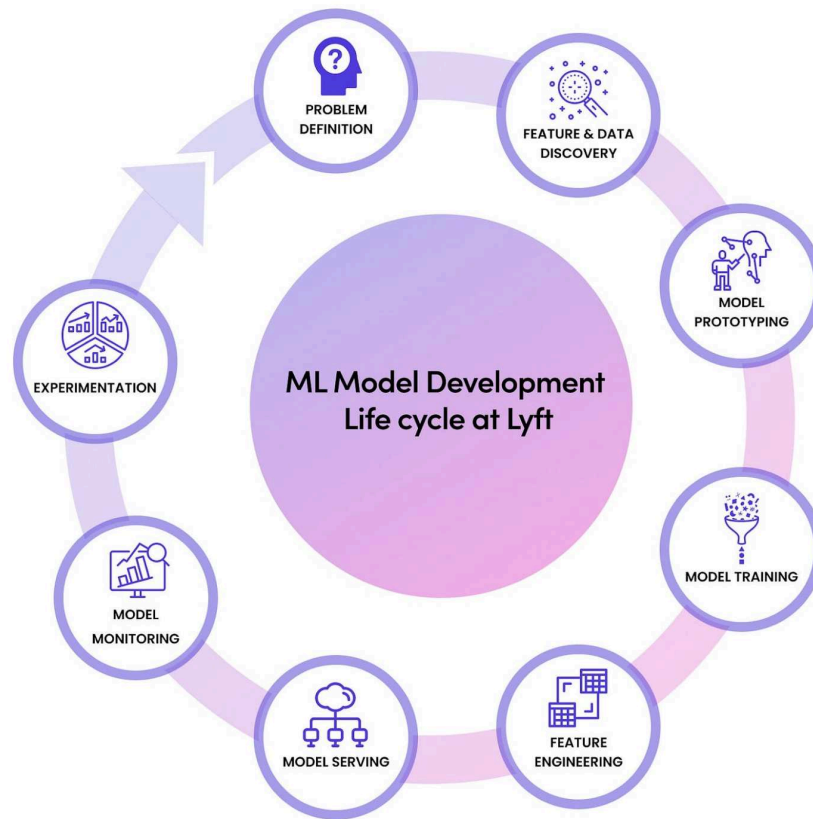
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DATA SCIENTIST JOB - REALITY

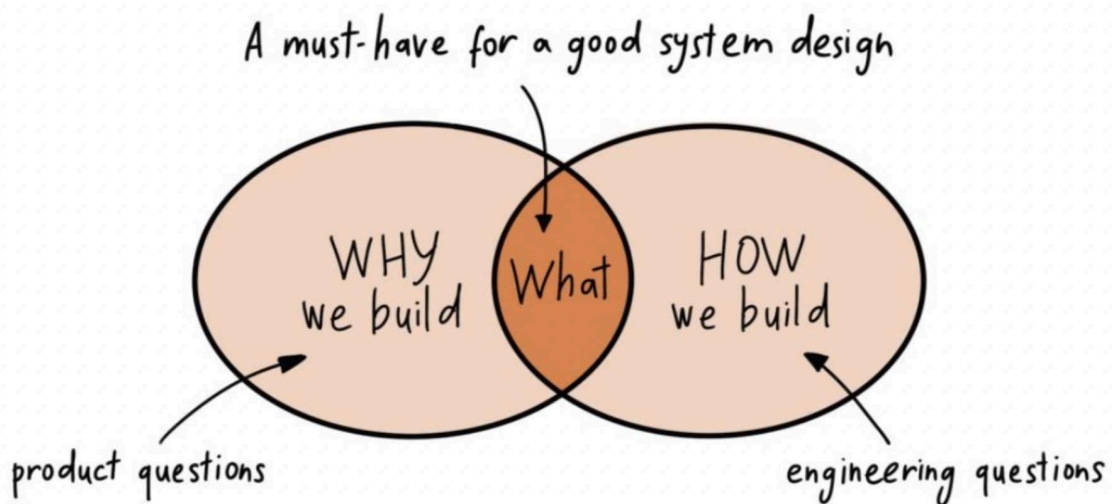


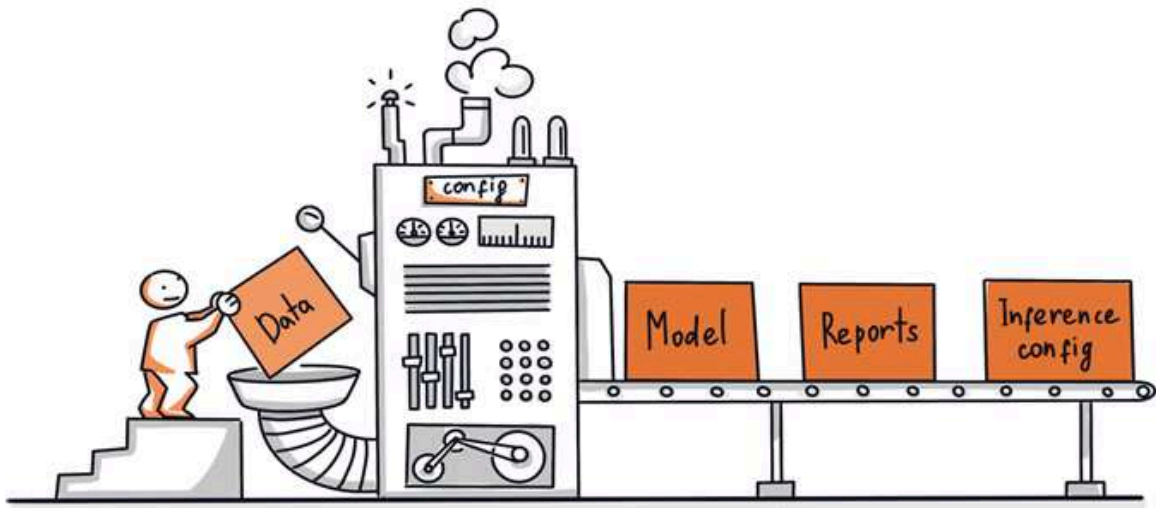
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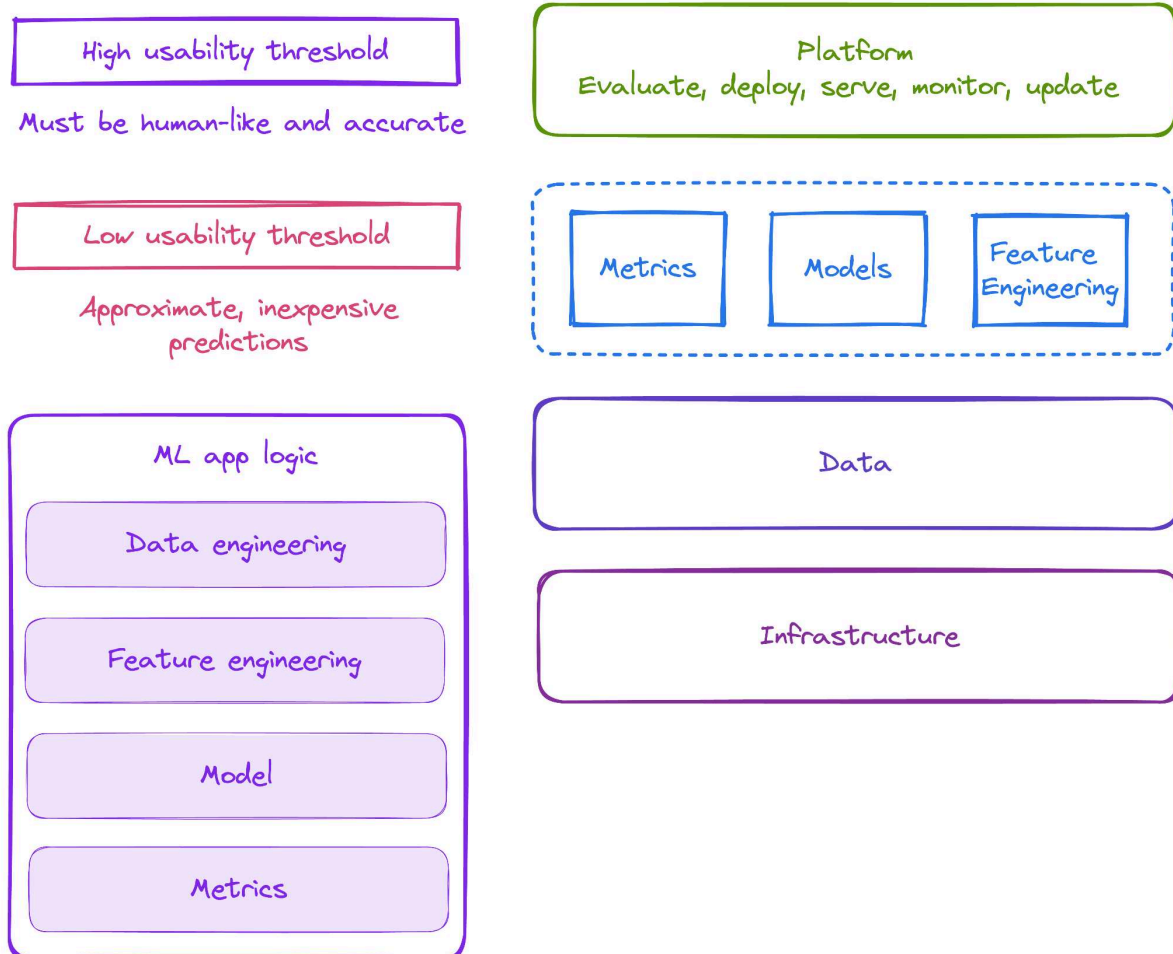


Illustrated ML Model Development Flow at Lyft

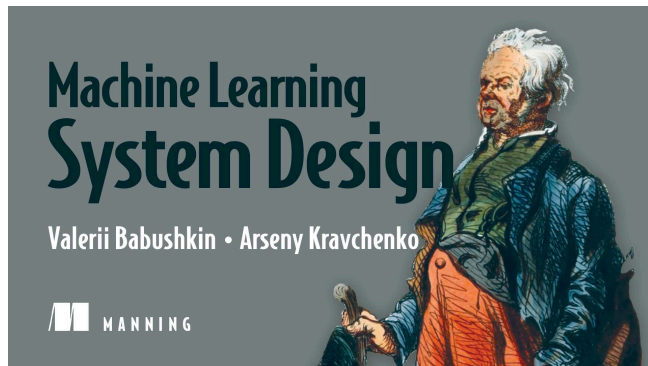




ML System Design



Notes from Book Machine Learning System Design by V. Babushkin and A. Kravchenko ↴



MACHINE LEARNING SYSTEM DESIGN is a complex, multistep process of *designing, implementing, and maintaining* machine learning-based systems that involves a combination of techniques and skills from various fields and roles, including machine learning, deep learning, software engineering, project management, product management, and leadership.

Machine learning experts get hired for a reason: companies need them to build, maintain, and improve systems—and not just for the sake of writing some code or closing Jira tickets. Businesses need reliable ML systems to reach objectives and solve problems.

ML system design is an architectural approach to engineering ML systems that incorporates experiences of hundreds of experts who worked in dozens of companies on a multitude of projects.

1. Why do we need to build the solution?
2. What problem does it solve?
3. Why does the problem occur?
4. What are the alternatives we know?
5. Why do we want to make it work with given limitations (metrics, latency, number of training samples)?

This design document blueprint will help us decide if we need to go further or it is better to stop right now and avoid a complicated ML solution.

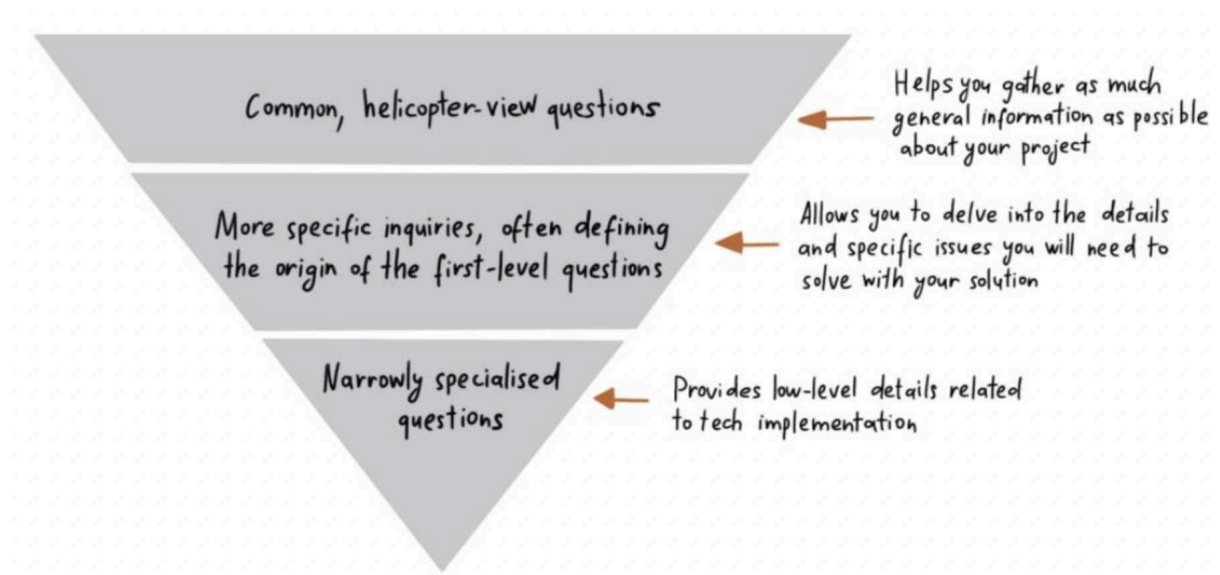
6. Why do we do this project?
7. How do we do it?
8. Do we have everything we need?
9. Can we have a less efficient solution with less effort?

- I. Problem definition
 - i. Origin
 - ii. Relevance & Reasons

- ii.i. Existing flow analysis
- ii.ii. Cost of not doing anything?
- ii.iii. Other reasons
- iii. Previous work
- iv. Other issues & Risks

Understand what problem we want to solve and gather context

Functional requirements determine what we should design, and non-functional requirements shape the understanding of how it should work under the hood

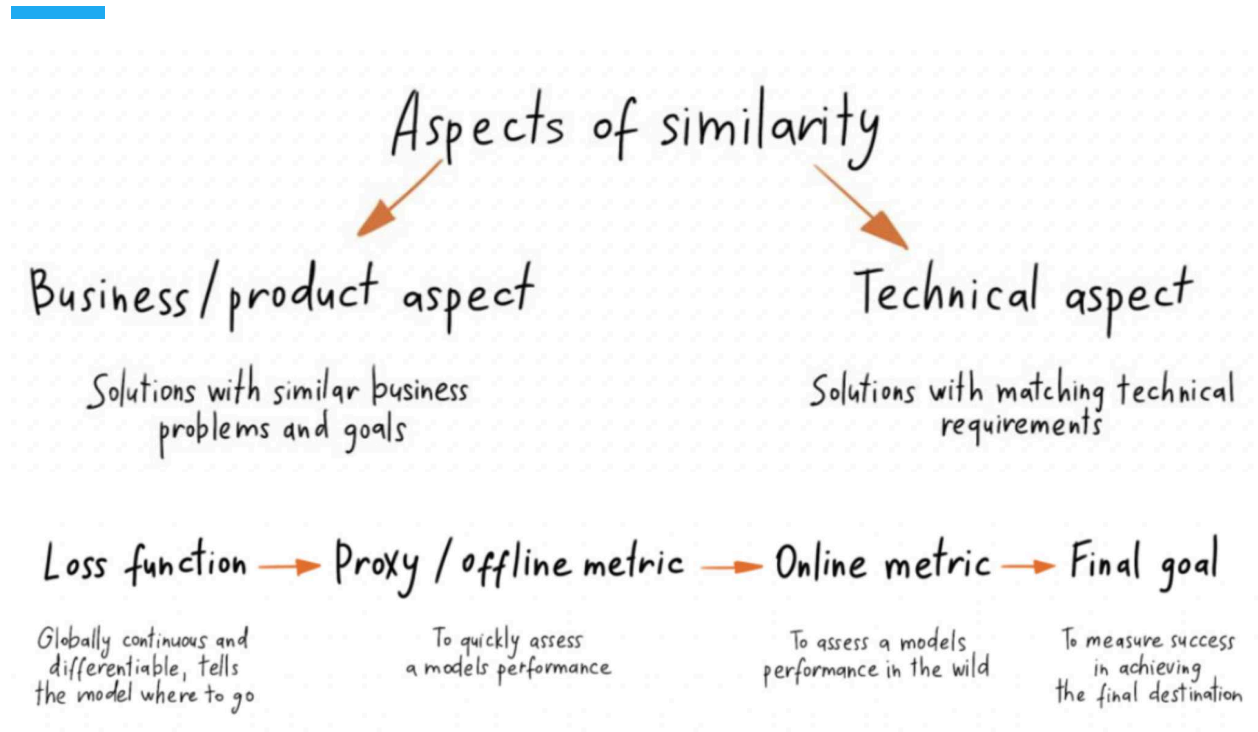


Identifying the risk landscape helps us understand what kind of problems are to be avoided. Some errors are almost harmless, some can affect business a lot, and some are life threats. Proper understanding of the costs of a mistake with regards to the system being designed is critical for the next steps, as it shapes requirements for reliability and data gathering, suggests better metrics, and may affect other aspects of design.

Is the problem related to the core part of the business?

End-to-end single model approach

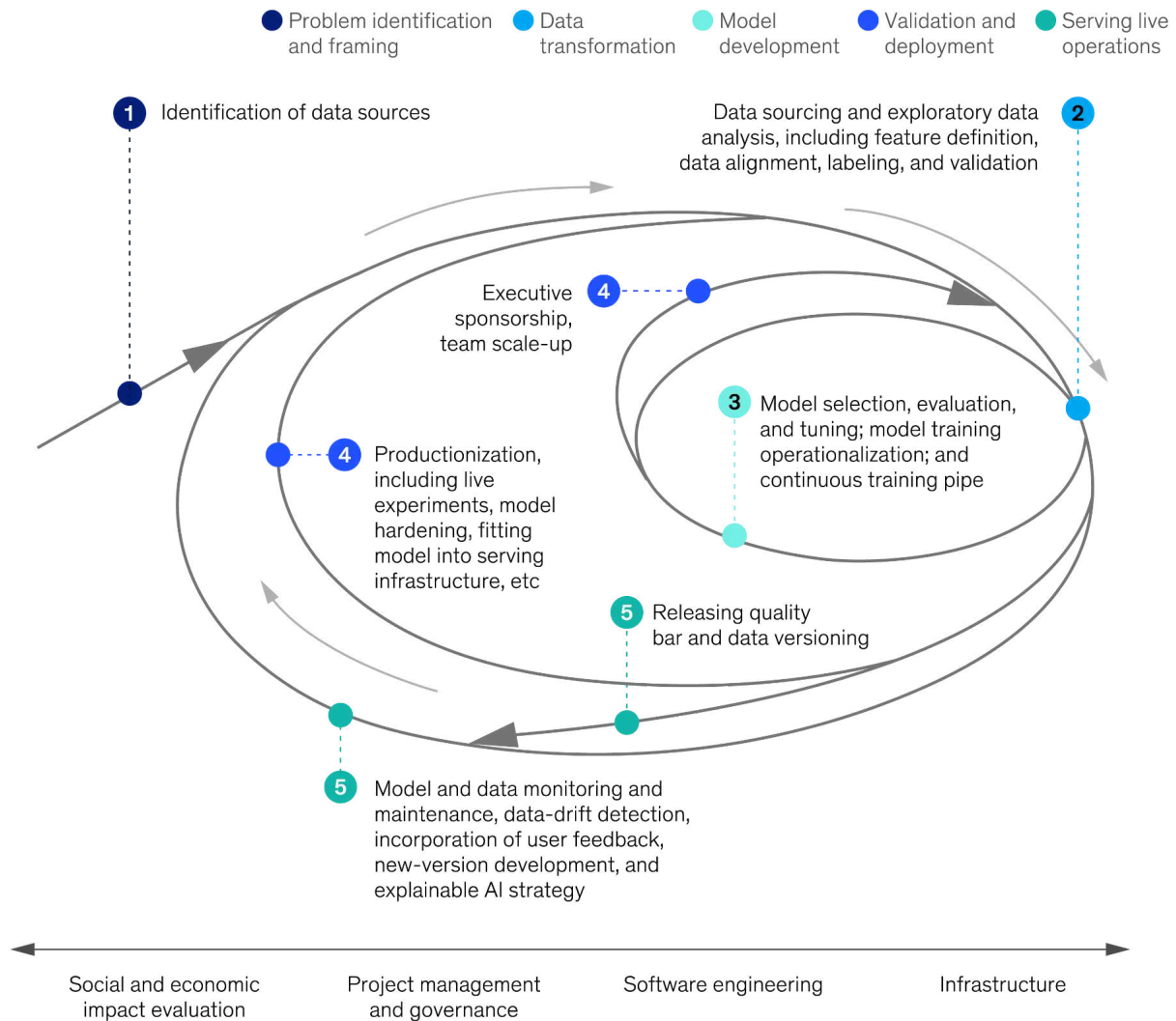
Minimum viable ML system → Average human-level ML system → Best-in-class ML system



- [CS 329S. Lecture 1. Understanding Machine Learning Systems in Production](#)
- [MotherDuck: Big Data is Dead; Big Data Is Dead... Long Live Big Data; A Guide to Data Roles](#)
- [SAI #28: Organisational structure for effective MLOps](#)
- [Real-time machine learning: challenges and solutions](#)
- [Building LLM applications for production](#)
- [Perspective: Unlocking ML requires an ecosystem approach | MLCommons](#)
- [Graph Machine Learning: An Overview | by Zach Blumenfeld | Apr, 2023 | Towards Data Science](#)
- [The Post-Modern Stack. Joining the modern data stack and the... | by Jacopo Tagliabue | Towards Data Science](#)

The ML development flywheel depicts the iterative process of creating an ML solution.

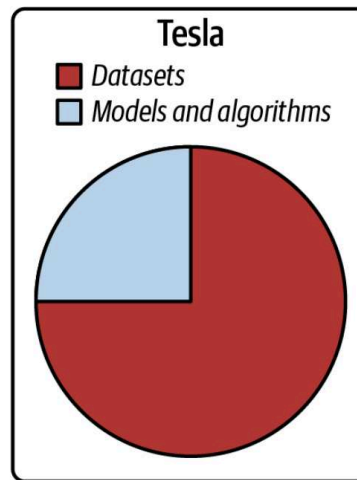
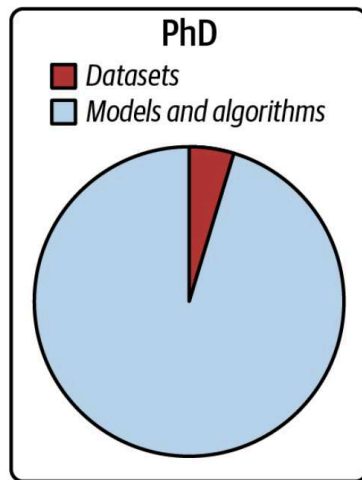
Life cycle of machine learning development

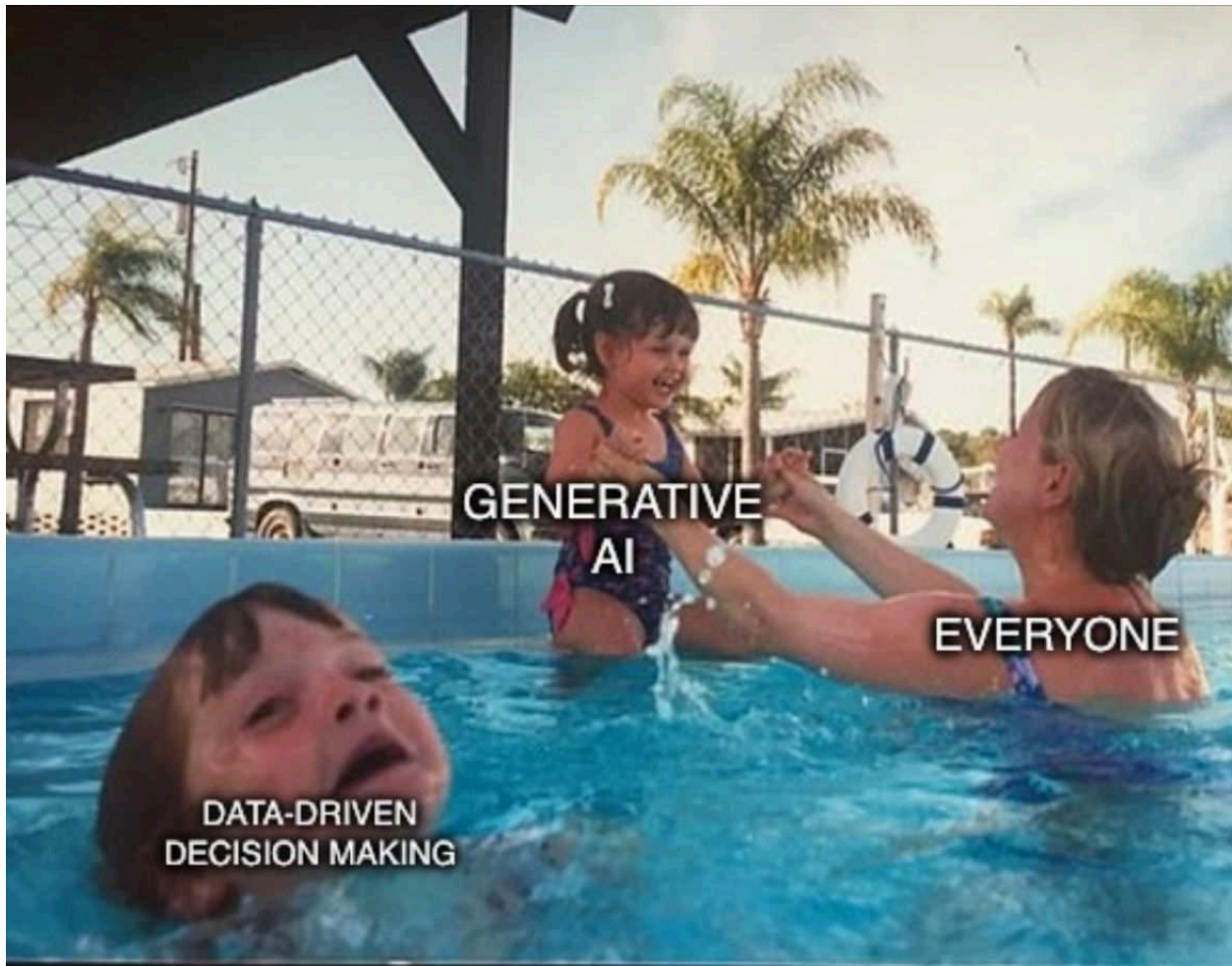


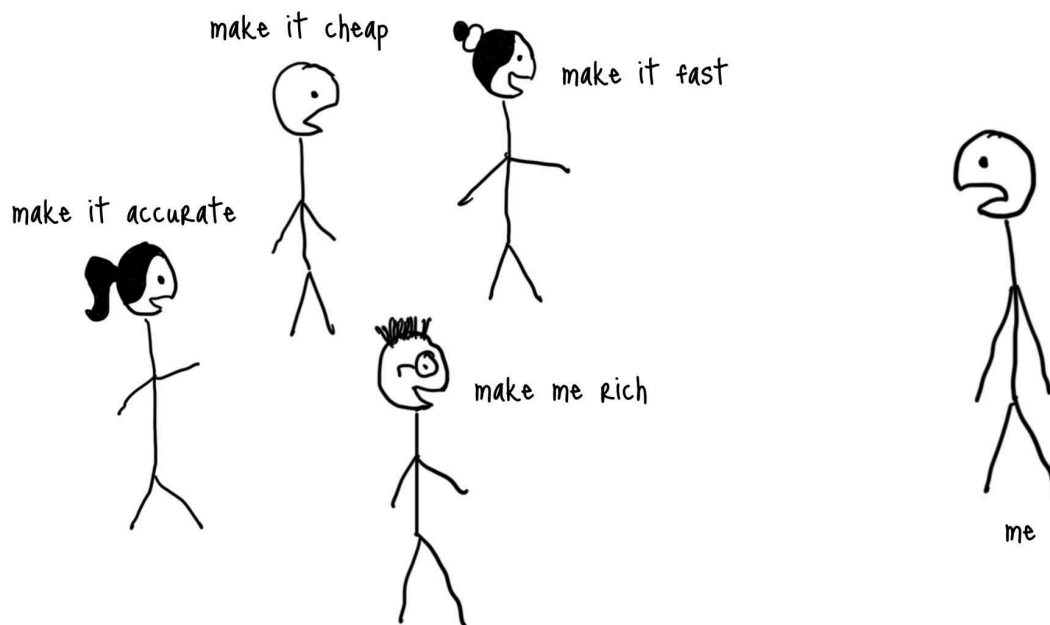
Source: Peter Mattson et al., "Perspective: Unlocking ML requires an ecosystem approach," MLCommons, March 10, 2023



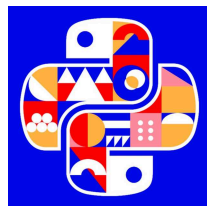
Amount of sleep lost over...







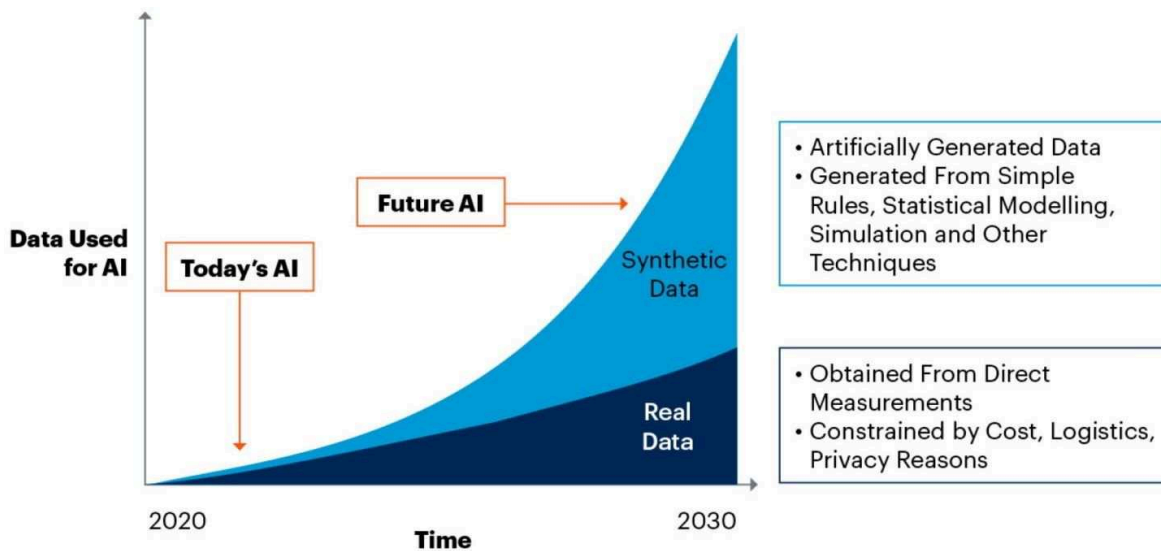
Python: Blogs and Resources



- [Dive into codebase like a pro](#)
- [Packaging Python Apps with Briefcase](#)
- [Responding to Earthquakes using Machine Learning and Racing through Time](#)
- [Fish and chips and Apache Kafka®](#)
- [Unleashing the Power of dbt and Python for Modern Data Stack](#)
- [Kraken Technologies: How we organise our very large Python monolith](#)
- [Python: practical uses for &=. I learnt something new today and... | by Pete Fison | Jul, 2023 | Medium](#)
- [GitHub - adriangonz/sigstore-talk: Supply Chain Security in MLSecOps](#)
- [Scipp](#)

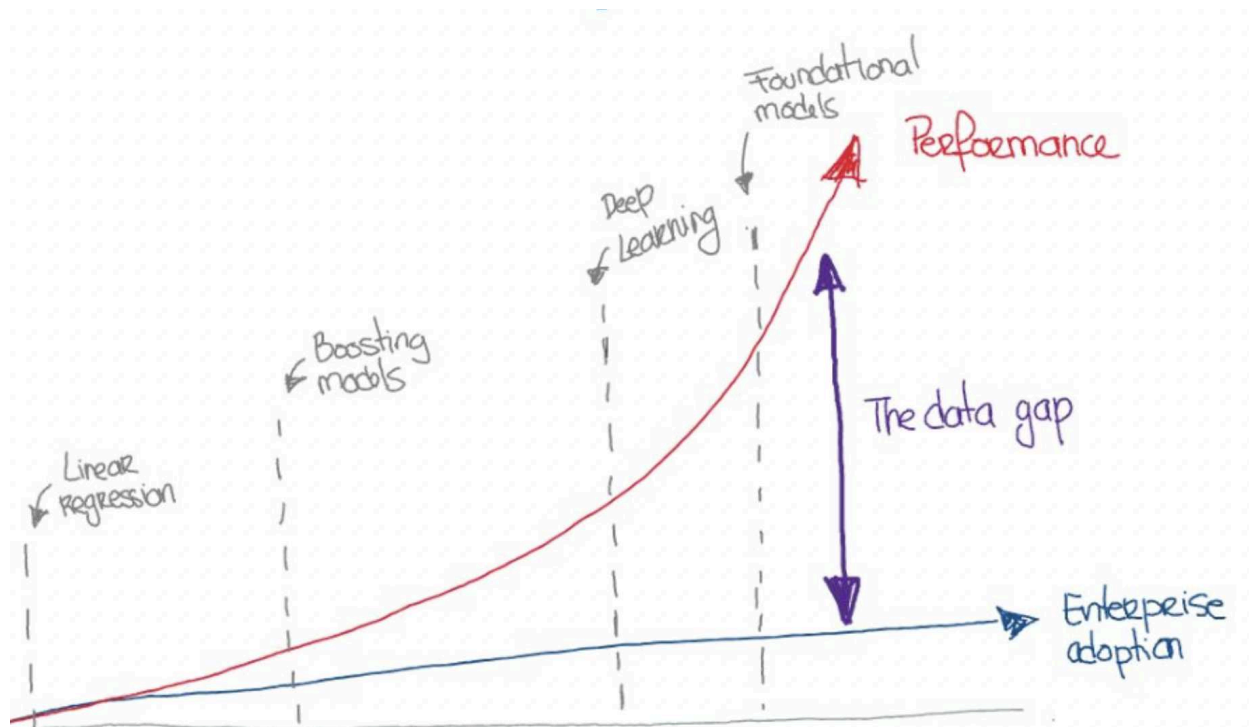
- Episode #165: Leveraging the Features of Your Database With Postgres and Python – The Real Python Podcast
- PostgreSQL: Documentation: 15: 8.17. Range Types
- this course!
- Craig Kerstiens
- GitHub - eli64s/readme-ai: 🚀 Generate beautiful README.md files from the terminal. Powered by OpenAI's GPT LLMs ✨
- Scaling scikit-learn - presented by Julien Jerphanion
- Hidden Features of Python - Stack Abuse
- Understanding Python Decorators and How to Use Them Effectively — Soshace
- Multiple assignment and tuple unpacking improve Python code readability
- FEA add TunedThresholdClassifier meta-estimator to post-tune the cut-off threshold #26120
- Python Decorators: Junior vs. Intermediate vs. Senior vs. Expert | by Gabe Araujo, M.Sc.
- Pandas Illustrated: The Definitive Visual Guide to Pandas | by Lev Maximov | Better Programming
- Building a blog in Django | Simon Willison's TILs
- One Liners Python Edition | Muhammad
- Traps and Pitfalls of Using SQL with Jinja
- Real-world match/case | Ned Batchelder
- Pandas (code snippet formatting!)
- Python: Just write SQL
- Boring Python: dependency management
- Python for Finance
- Your Friendly Guide to Colors in Data Visualisation
- Efficient Python Tricks and Tools for Data Scientists
- HostJupyter - Best Jupyter Book Alternative
- Text and Annotation | Python Data Science Handbook
- What is a data artist? A role for drawing IT and business together | CIO

Synthetic Data



Source: Gartner
750175_C

Experian report: 63% prioritizing investments in AI-generated synthetic data to accelerate their analytics development cycle. With these enhanced datasets, institutions can refine their credit risk models, improve accuracy in predicting defaults, and tailor their offerings to cater to a broader range of clientele. Moreover, the continuous evolution and integration of synthetic data can foster innovation and drive the industry toward more inclusive and adaptive financial solutions.



Combining Great Expectations with Fabric to create better datasets for Machine Learning

Generate synthetic data given AUC (Copy of original variable based on ROC AUC curvature)
 → transform using known summary stats ↴

26.2.23

$$x_p = t - \frac{c_0 + c_1 t + c_2 t^2}{1 + d_1 t + d_2 t^2 + d_3 t^3} + \epsilon(p), \quad t = \sqrt{\ln \frac{1}{p^2}}$$

$$|\epsilon(p)| < 4.5 \times 10^{-4}$$

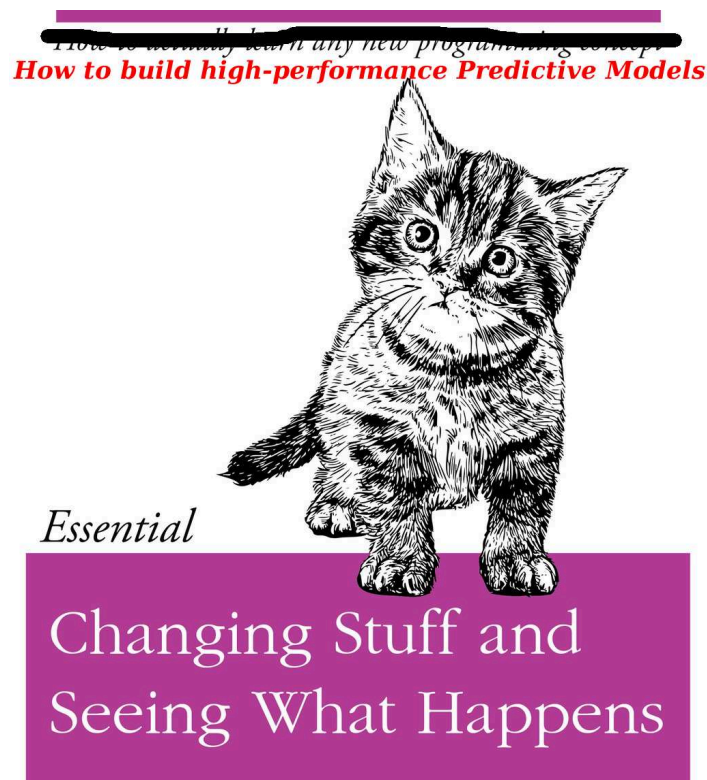
$$c_0 = 2.515517 \quad d_1 = 1.432788$$

$$c_1 = .802853 \quad d_2 = .189269$$

$$c_2 = .010328 \quad d_3 = .001308$$

- How to simulate a calibrated prediction model given prevalence and auc - Cross Validated
- Generate synthetic data given AUC - Cross Validated
- How to generate synthetic data for machine learning projects | Openlayer
- How to generate synthetic data from real data - zero to hero - MOSTLY AI
- fast.ai - How (and why) to create a good validation set
- Exploratory Undersampling for Class-Imbalance Learning
- Are unbalanced datasets problematic, and (how) does oversampling (purport to) help? - Cross Validated

MLOps / ML Engineering



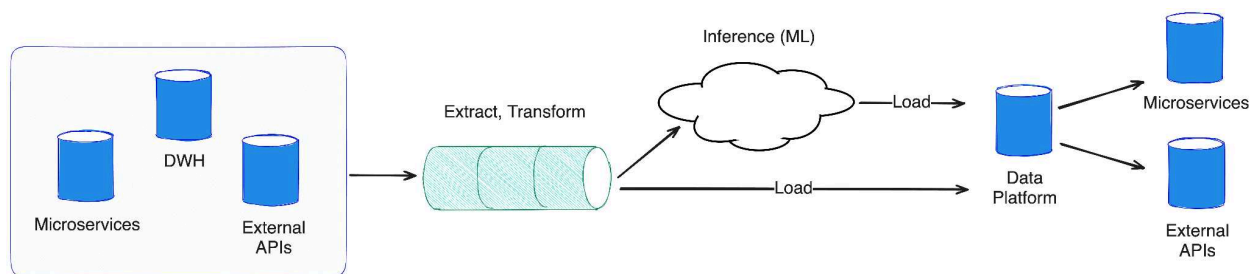
O RLY?

@ThePracticalDev

- MLOps Principles

- [CS 329S | Syllabus](#)
- [661: Designing Machine Learning Systems — with Chip Huyen](#)
- [cs329s_2022_08_slides_deployment](#)
- [SAI #23: Deconstructing a Feature Store](#)
- [Tecton](#)
- [Reading and Writing the Apache Parquet Format](#)
- [One SQL to Rule Them All](#)
- [Building an ML Build System from scratch using only Python and Docker | Qwak's Blog](#)
- [MLOps is 98% Data Engineering](#)
- [Self-Serve Apps for ML Teams | Greylock](#)
- [MLOps Zoomcamp 2.4 - Model management](#)
- [Explain Model Outcomes, Root Out Model Bias - Arize AI](#)
- [The MLOps Stack](#)
- [Can data science survive the rise of AutoML? - Quora](#)
- [Deep learning enhancing banking services: a hybrid transaction classification and cash flow prediction approach - PMC](#)
- [Dealing with Train-serve Skew in Real-time ML Models: A Short Guide - Building Nubank](#)
- [Notes for CS181: Machine Learning](#)
- [Introduction to Microservices, Docker, and Kubernetes](#)
- [How to Deploy a Machine Learning Model to Google Cloud for 20% Software En...](#)
- [How to be a Good Machine Learning PM by Google Product Manager, Rubén Loz...](#)
- [3 Common Problems With Your Machine Learning Product and How to Fix Them](#)
- [Machines Can See 2018](#)
- [CS 329S: Machine Learning Systems Design](#)
- [Feature Stores for Realtime Machine Learning - What, Why, How](#)
- [Using MLOps to Build a Real-time End-to-End Machine Learning Pipeline | Binance Blog](#)
- [Rules of Machine Learning: | Google Developers](#)
- [MLOps Principles](#)
- [The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction](#)
- [Feature Store - Made With ML](#)

- Real-time machine learning: challenges and solutions
- Data Curious



Notes from [\[DET Webinar\] Career Growth in Data Engineering with Shachar Meir](#) ↗

Technical part is the easiest part to learn. The tools all do exactly the same thing. Data engineering → don't focus much on the business aspects, learning business. FinTech: Risk and Fraud. Spending time with business people. Tech is important only in 1st and 2nd year, it's not what's gonna help you grow.

Unless you're a software engineer → you need to prove your worth (e.g., Data Scientist). Talk about your work in a way that's inspiring for people.

Architecture is about soft skills, putting together a vision, to believe that's the right way of doing things. A leadership role, you're not managing people but setting direction and having people follow you. It's not just about technology.

Amazing team stack description ↗

Our primary language is Python, our core platform consists of a Snowflake data warehouse, fueled primarily by an on site Airflow instance running on a Kubernetes cluster. Data is centralized through custom orchestrated batch jobs, with S3 as our intermediary lake. Data modelization is enabled thanks to an orchestrated dbt repository and quality validation results of a mix of tests using Great Expectations and dbt. We took advantage of ELK for logging and monitoring as well as Airflow's internal database.

Testing Strategies

Unit Testing

Testing individual components (data pre-processing, feature extraction et al.)

Integration Testing

Verifying the interactions and compatibility b/w various components (data pipeline)

Model Testing

Examine the performance and the behavior of ML model (evaluating metrics like PRF scores)

Robustness Testing

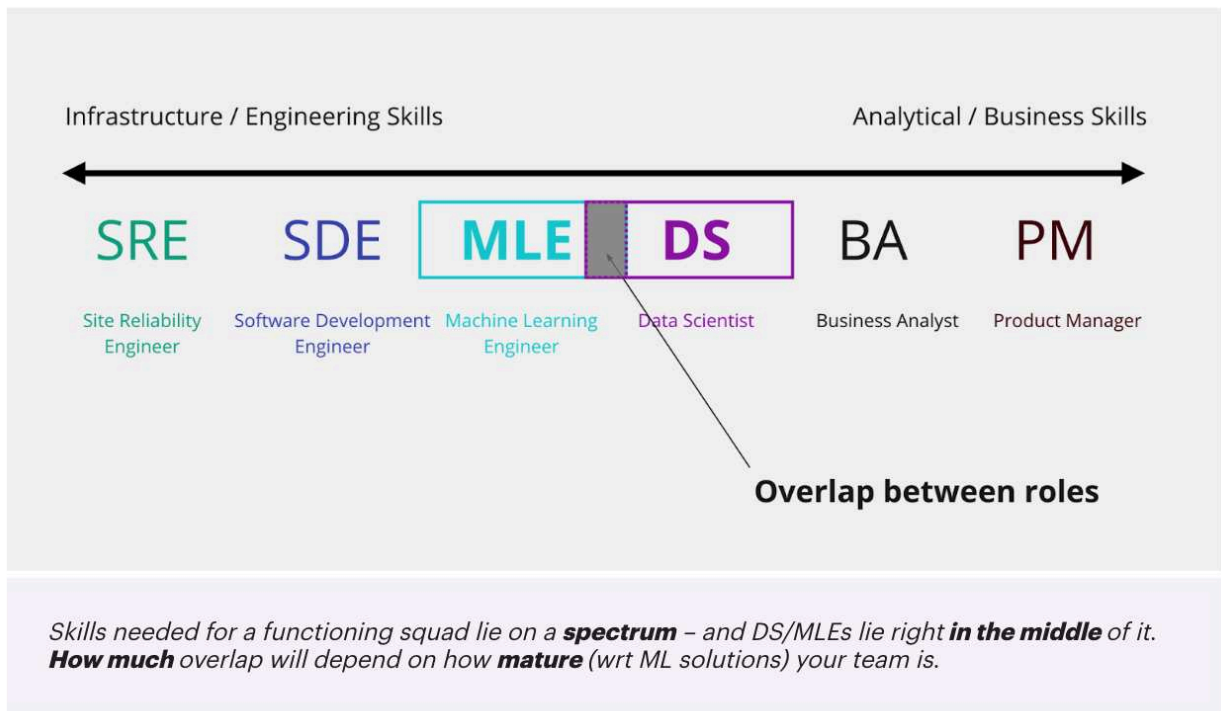
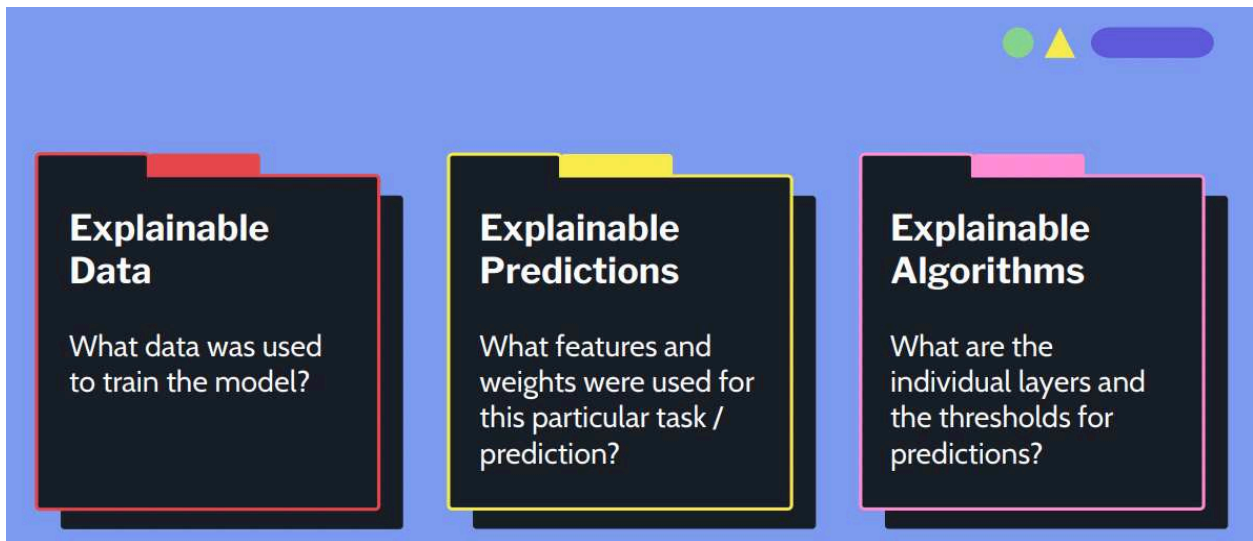
Examine the system's ability to handle edge cases, outliers and adversarial inputs.

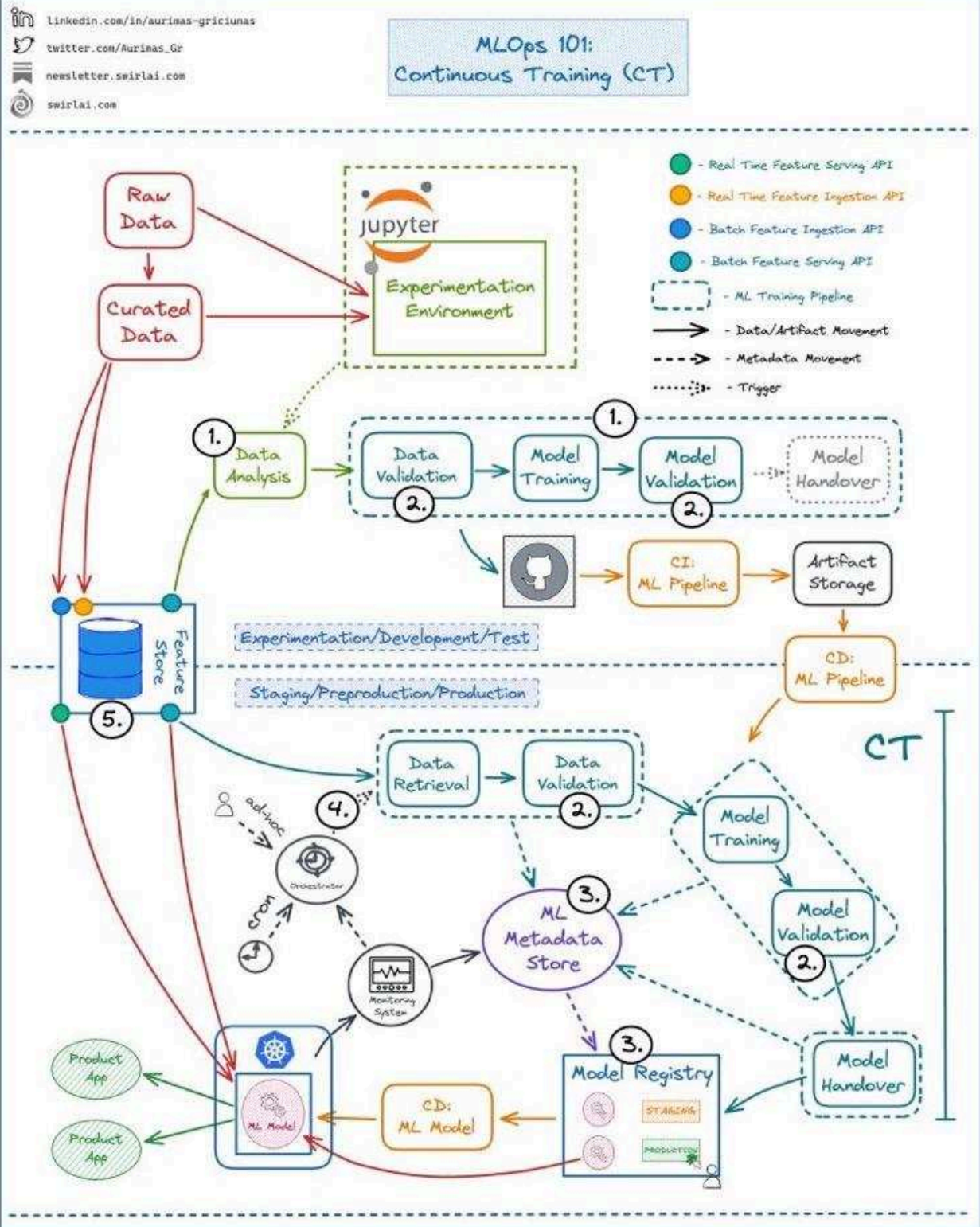
Deployment Testing

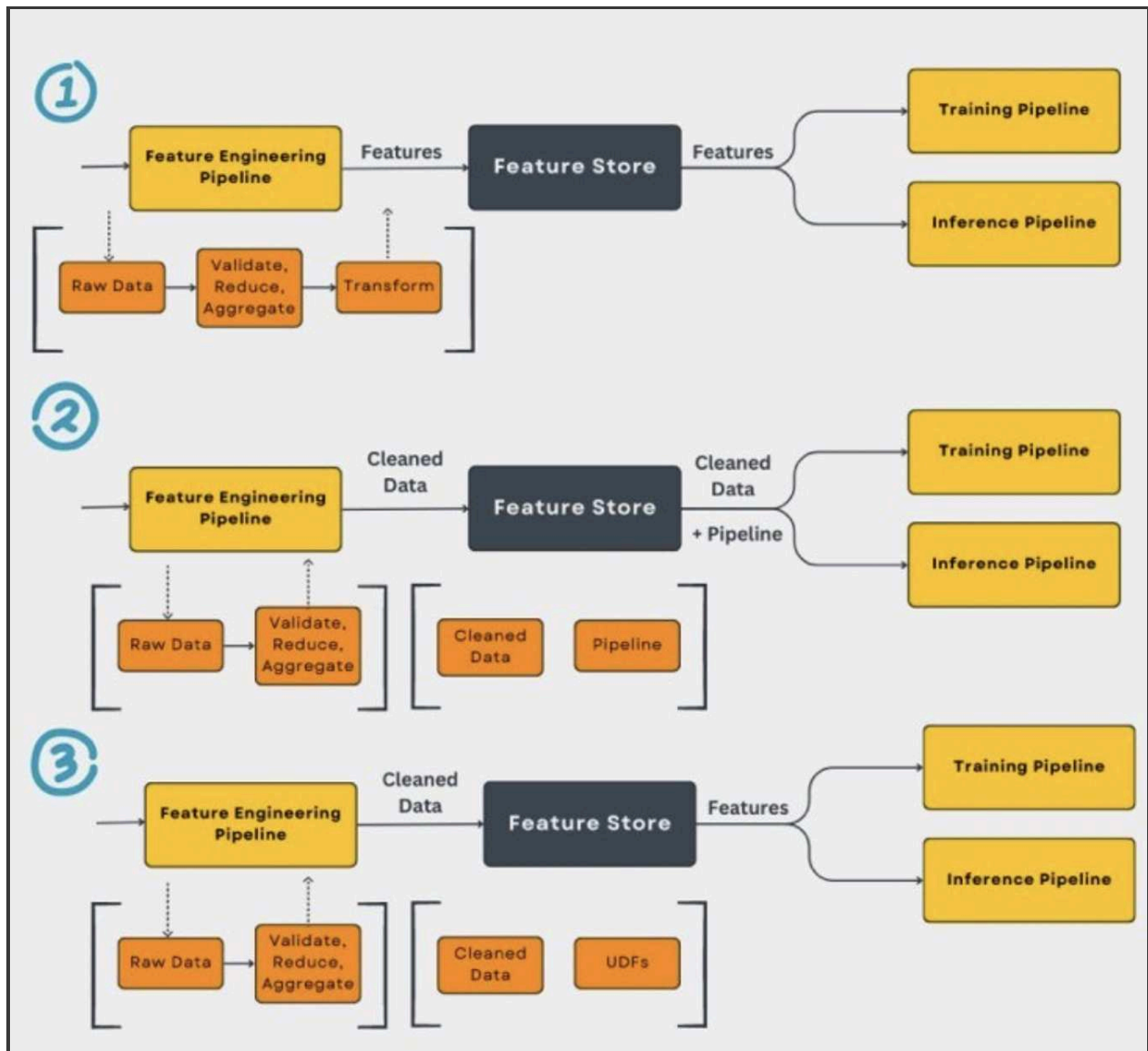
Validating ML system in production environment under real-world conditions

Ethical & Fairness Testing

Evaluation for bias, discrimination & adherence to ethical regulations









linkedin.com/in/aurimas-griciunas



twitter.com/Aurimas_Gr



newsletter.swirlai.com

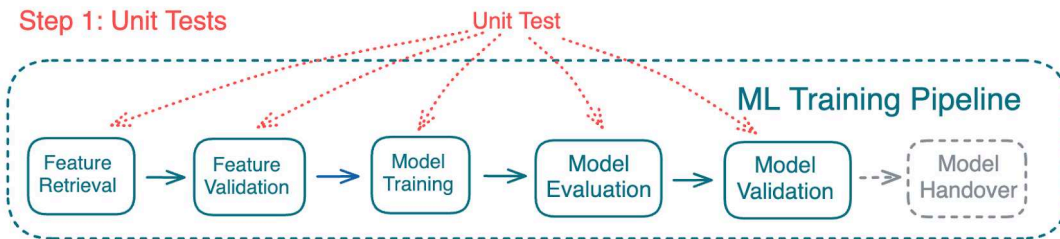


swirlai.com

CI/CD for Machine Learning

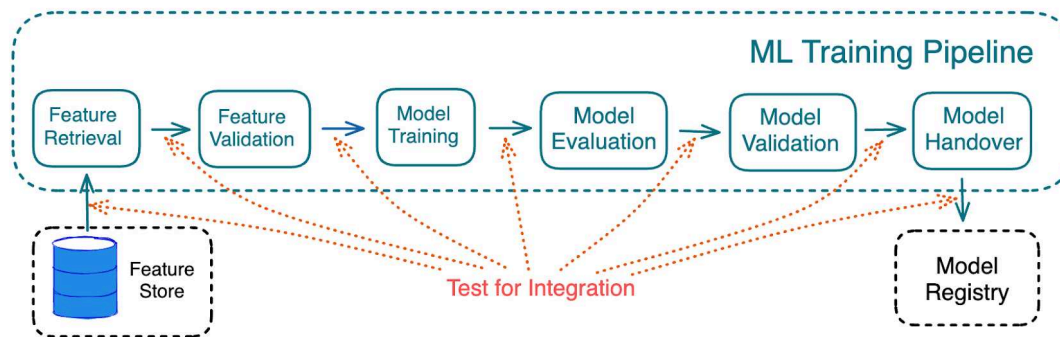
CI/CD Pipeline

Step 1: Unit Tests



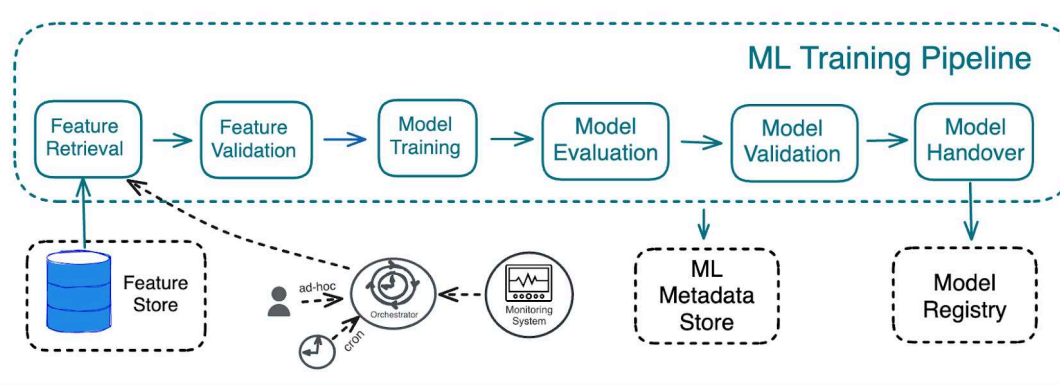
Step 2: Integration Tests

Pre-Prod environment



Step 3: Delivery

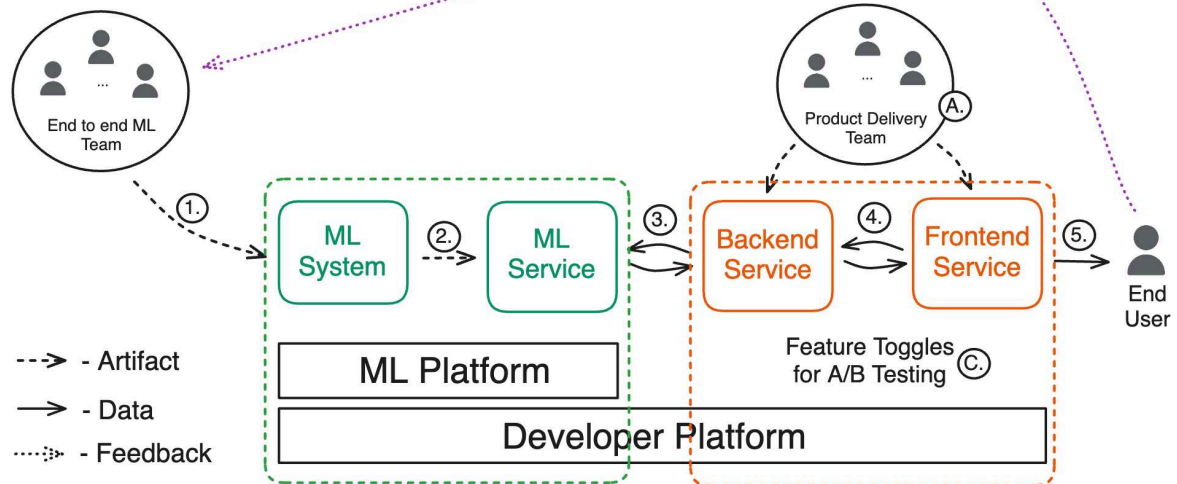
Pre-Prod/Prod environment



ML Systems in the context of end-to-end Product Delivery

Responsibility Boundaries

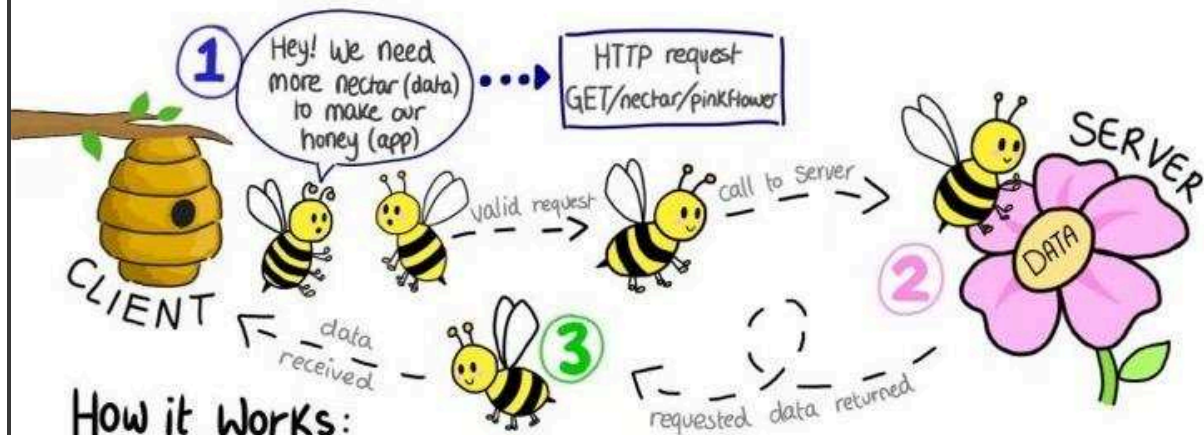
- End-to-end ML Team
- Product Delivery Team



What is an API?

@Rapid_API 

An application programming interface allows two programs to communicate. On the web, APIs sit between an application and a web server, and facilitate the transfer of data.



How it Works:

1 Request

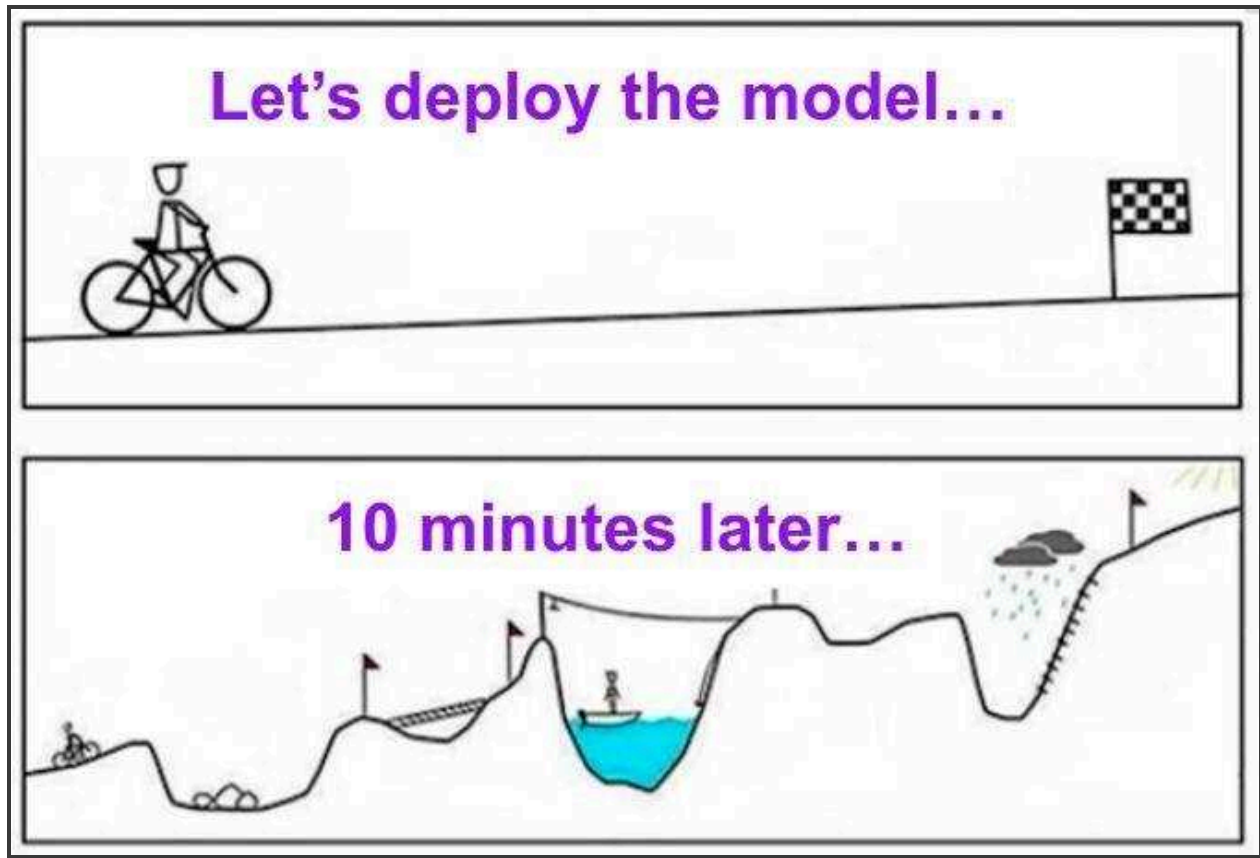
API call is initiated by the Client application via a HTTP request

2 Receive

Our worker bee acts as an API, going to a flower (server) to collect nectar (data)

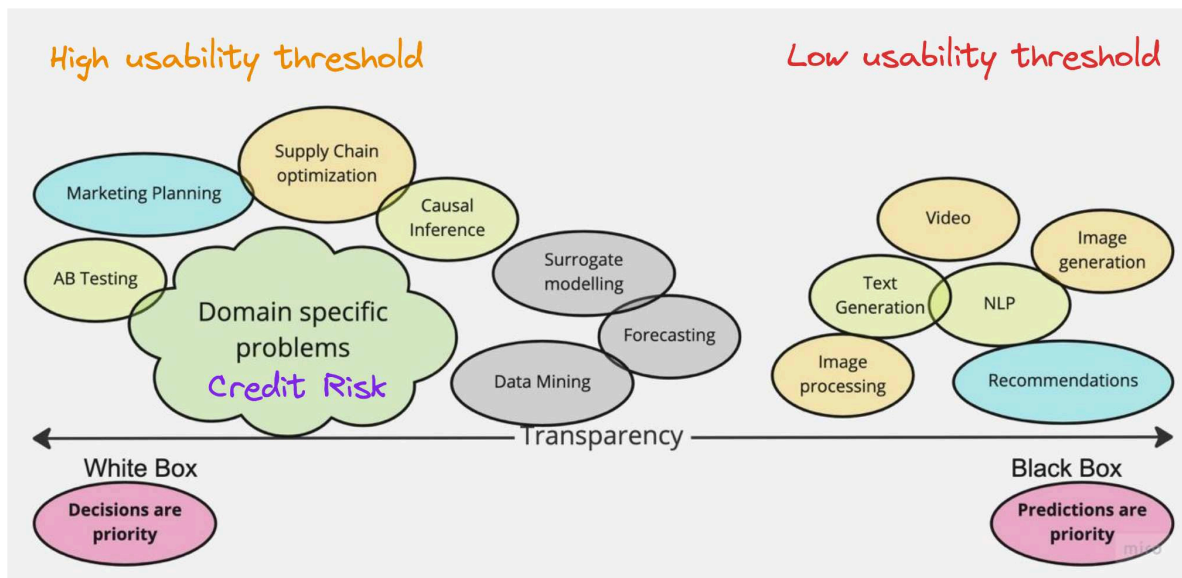
3 Response

The API transfers the requested data back to the requesting application, usually in JSON format



Model Risk Management

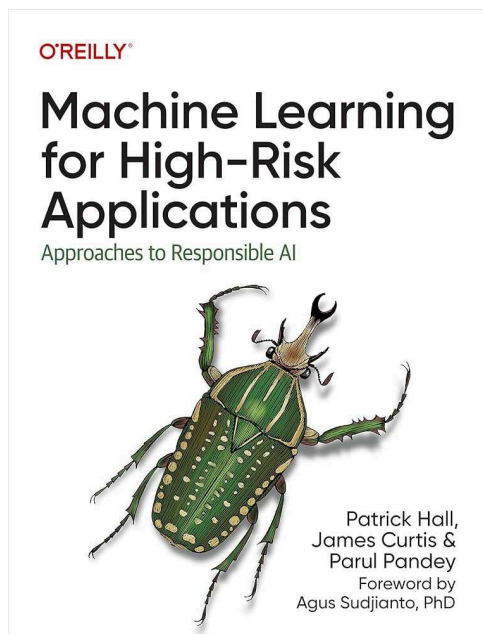
- [What Is AI Model Governance?](#)
- [Regulating AI Innovation: Aboitiz Data Innovation's David Hardoon](#)
- [Fiddler Auditor is a tool to evaluate the robustness of language models.](#)



What is Model Critique?

If the model fails any of the above four evaluation steps, we need to revisit and potentially re-do each design decision:

1. Is our training data the "right" data set?
2. Is the functional form (linear or polynomial of degree 2) we assumed correct?
3. Does our loss function capture what we really want to happen in real-life?
4. Did we optimize well?
5. Did we provide the right information about the model to the human decision maker?



Notes from Patrick Hall “Machine Learning for High-Risk Applications” ↴

Written Policies and Procedures: The organizational rules for making and using ML should be written and available for all organizational stakeholders. Those close to ML systems should have trainings on the policies and procedures. **These rules should also be audited to understand when they need to be updated.** No one should be able to claim ignorance of the rules, the rules should be transparent, and the rules should not change without approval. Policies and procedures should include clear mechanisms for escalating serious risks or problems to senior management, and likely put forward whistleblower processes and protections.

Effective Challenge: Effective challenge dictates that experts with the capability to change a system, who did not build the ML system being challenged, perform validation and auditing. MRM practices typically distribute effective challenge across three “lines of defense,” where **conscientious system developers make up the first line of defense** and independent, skilled and empowered **technical validators** and **process auditors** make up the second and third lines, respectively.

Accountable Leadership: A specific executive within an organization should be accountable for ensuring AI incidents do not happen. This position is often referred to as **Chief Model Risk Officer (CMRO)**. It’s also not uncommon for CMRO terms of employment and compensation to be linked to **ML system performance**. The role of CMRO offers a very straightforward cultural check on **ML safety and performance**. If our boss really cares about ML system safety and performance, then we start to care too.

Incentives: Data Science staff and management must be **incentivized to implement ML responsibly**. Often, compressed product timelines can incentivize the creation of a minimum viable product first, with rigorous testing and remediation relegated to the end of the model life cycle immediately before deployment to production. Moreover, ML testing and validation teams are often evaluated by the same criterion as ML development teams, leading to a fundamental misalignment where testers and validators are encouraged to move quickly rather than assure quality. Aligning timeline, performance evaluation, and pay incentives to team function helps solidify a culture of responsible ML and risk mitigation.

Of course, small or young organizations may not be able to spare an entire full-time employee to monitor **ML system risk**. But it’s important to have an individual or group held accountable if ML systems cause incidents and rewarded if the systems work well. **If an organization assumes everyone is accountable for ML risk and AI incidents, the reality is that no one is accountable.**

Participating in the design and implementation of high-impact AI systems requires a mindset change to prevent egregious performance and safety problems. Practitioners must change from prioritizing the number of software features they can push or the test data accuracy of an ML model, to recognizing the implications and downstream risks of their work

The product of risk and loss has a more formal name in the context of MRM, materiality

Adversarial mindset, or at least to block out overly positive ML hype.

Many machine learning workflows are still centered around black box models. Such black box models are a common type of, often unnecessary, complexity in a commercial ML workflow

Model assertions, also known as business rules, are a mitigation technique that can be applied directly to ML model predictions to address such discovered failure modes.

Python

```
def get_monotone_constraints(data, target, corr_threshold):
    corr = pd.Series(data.corr(method="spearman")[target]).drop(target)
    monotone_constraints = tuple(
        np.where(corr < -corr_threshold, -1, np.where(corr > corr_threshold, 1,
    0))
    )
    return monotone_constraints
```

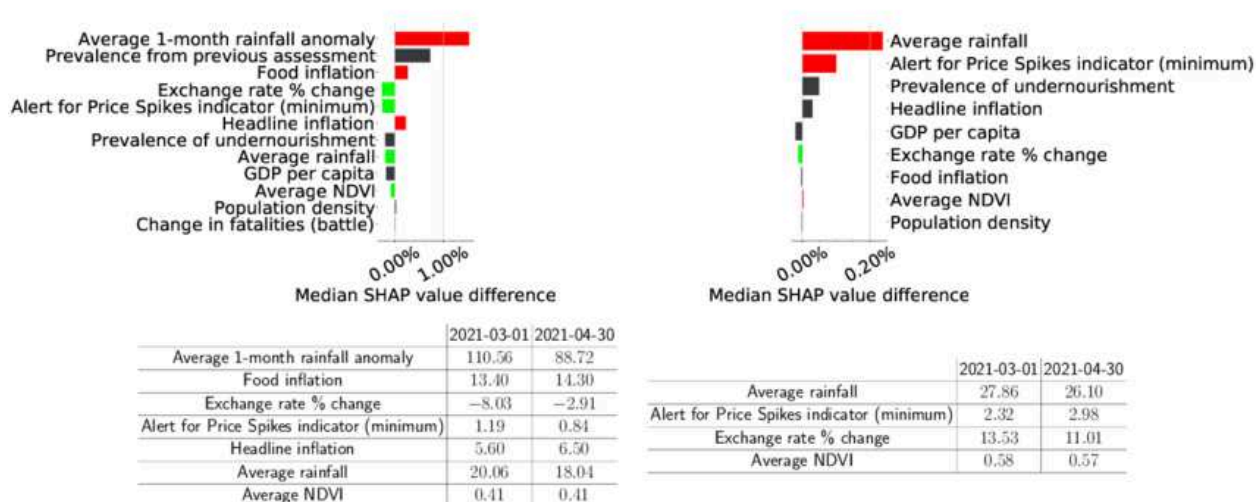
- [Preventing Repeated Real World AI Failures by Cataloging Incidents: The AI Incident Database](#)
- [Advancing accountability in AI : Governing and managing risks throughout the lifecycle for trustworthy AI | OECD Digital Economy Papers](#)
- [Exploring Tools for Interpretable Machine Learning - Dr. Juan Camilo Orduz](#)
- [Fiddler Report Generator for AI Risk and Governance](#)

Data Privacy

- [Is Credit Scoring an Automated Decision? – The Opinion of the AG Pikamäe in the Case C-634/21 — The Digital Constitutionalist](#)
- [Majority of credit bureau "CRIF" database illegal](#)
- [\[BaFin\] When an algorithm decides on the loan](#)

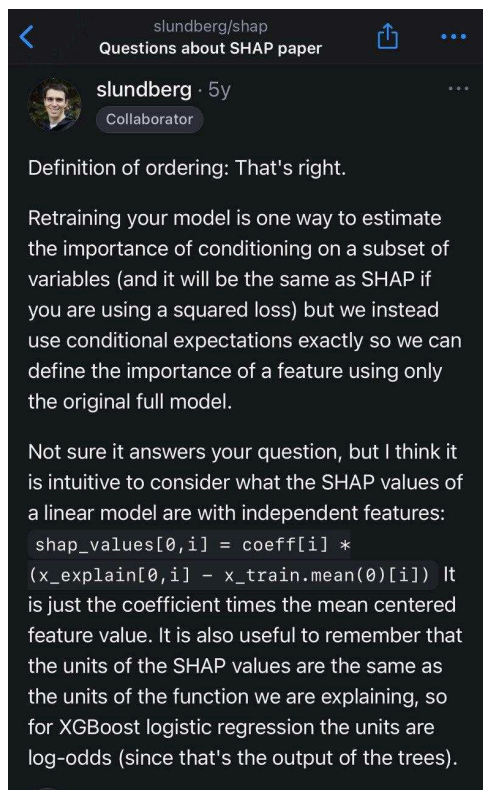
- [Proposal for a regulation of the European Parliament and of the Council on harmonised rules on Artificial Intelligence \(Artificial Intelligence Act\) and amending certain Union Legislative Acts](#)
- [Experian appeals the ICO's Enforcement Notice – who comes out best? - BDB Pitman](#)
- [CJEU landmark rulings on “credit ranking” and review of DPAs](#)
- [Credit scoring firms face curbs after landmark EU data protection ruling | TechCrunch](#)
- [Face Detection - Google Cloud Model Cards](#)
- [GitHub - tensorflow/model-card-toolkit: A toolkit that streamlines and automates the generation of model cards](#)
- [A Python library to check the level of anonymity of a dataset | Scientific Data](#)

Model Explainability



- [9.6 SHAP \(SHapley Additive exPlanations\) | Interpretable Machine Learning](#)
- [An introduction to explainable AI with Shapley values — SHAP latest documentation](#)
- [How to Easily Customize SHAP Plots in Python | Towards Data Science](#)
- [SHAP: Shapley Additive Explanations | by Fernando López](#)
- [\(PDF\) Nowcasting food insecurity on a global scale](#)
- [lux · PyPI](#)
- [GitHub - SelfExplainPiML \(Python Interpretable Machine Learning\) toolbox for model development & diagnostics](#)
- [Data Exchange Podcast \(Episode 180\): Patrick Hall of BNH and Agus Sudjianto of ...](#)

- SHAP not working with LightGBM categorical features - Stack Overflow
- What's in the Box?: Cynthia Rudin's Quest to Solve AI's Black Box Problem | Impact | Giving to Duke
- AI Explained: Machine Learning for High Risk Applications
- GitHub - oegedijk/explainerdashboard: Quickly build Explainable AI dashboards that show the inner workings of so-called "blackbox" machine learning models.
- Interpretable machine learning for imbalanced credit scoring datasets - ScienceDirect
- Understanding Shapley value explanation algorithms for trees
- Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead
- We Didn't Explain the Black Box – We Replaced it with an Interpretable Model
- Duke Data Science Entry to FICO Data Challenge
- An Explainable Model for Credit Risk Performance
- Why the explainable AI market is growing rapidly | VentureBeat
- S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in neural information processing systems, pp. 4765–4774, 2017.
- Examples - Evidently Documentation
- GitHub - interpretml/interpret: Fit interpretable models. Explain blackbox machine learning
- Postprocessing - AIF360
- AI and housing discrimination: the case of mortgage applications | SpringerLink
-  E021: Explainable and interpretable AI, with Serg Masis
- GitHub - whylabs/whylogs: The open standard for data logging
- fasttreshap · PyPI
- How Shapley Values Work



ML Metrics

- [Statistical Thinking - Classification vs. Prediction](#)
- [Damage Caused by Classification Accuracy and Other Discontinuous Improper Accuracy Scoring Rules](#)
- [Why is accuracy not the best measure for assessing classification models? - Cross Validated](#)
- [In python, how to discretize continuous variable using accuracy as a criterion taking class into consideration - Stack Overflow](#)
- [Class-Balanced Loss Based on Effective Number of Samples](#)
- [Understanding binary cross-entropy / log loss: a visual explanation | by Daniel Godoy | Towards Data Science](#)
- [Occam razor vs. machine learning](#)
- [The predictive accuracy of credit ratings: Measurement and statistical inference Munich Personal RePEc Archive](#)

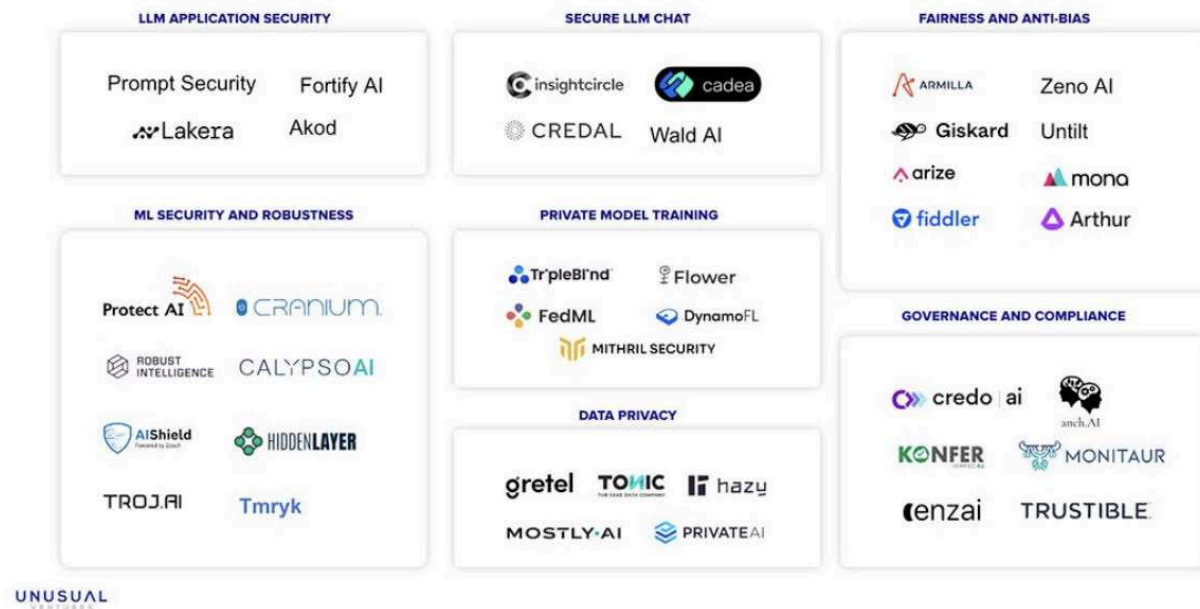
- [Using the Gini coefficient to evaluate the performance of credit score models | by Idan Schatz | Towards Data Science](#)

Calibration

- [Researchers Explore Machine Learning Calibration -- Pure AI](#)
- [prob_calibration notebook, calibrator.py](#)
- [Optimal clustering for PD calibration in the presence of noise, Alexander Marianski](#)
- [Polynomial generalisation of discrete PD calibration for continuous direct estimation](#)
- [Neural Network Calibration using PyTorch | by Lukas Huber | Towards Data Science](#)
- [Why model calibration matters and how to achieve it](#)
- [The confidence calibration problem in machine learning | Dasha.AI](#)
- [Estimating Expected Calibration Errors](#)
- [Calibration — Applied Machine Learning in Python](#)
- [Calibrating for Class Weights by Modeling Machine Learning](#)
- [Understanding Class Weights | Data Science and Machine Learning | Kaggle](#)
- [Is There a Way to Re-Calibrate Predicted Probabilities After Using Class Weights?](#)
- [Calibrating Probability with Undersampling for Unbalanced Classification](#)
- [Network calibration by weight scaling | OpenReview](#)
- [Confidence Calibration for Deep Networks: Why and How?](#)
- [Calibrating Deep Neural Networks using Focal Loss \(NeurIPS'20\)](#)
- [KDD 2020 | How to calibrate your neural network classifier: Getting accurate probabilities from a classification model](#)
- [Calibration alternatives to logistic regression \[Bundesbank\]](#)
- [pycalibrate is a Python library to visually analyze model calibration in Jupyter Notebooks](#)

Fairness

Responsible AI





Building fair, accountable, and trustworthy machine learning systems is critical for ensuring that the benefits of these systems are widely accessible and enjoyed by all.

By ensuring that our models are transparent and interpretable, we can detect and mitigate biases and unfairness before they cause harm.

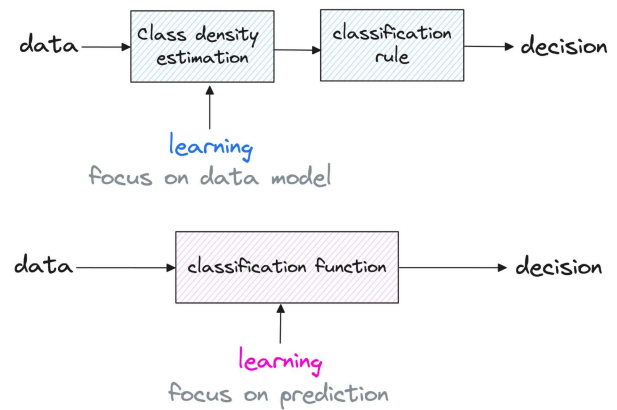
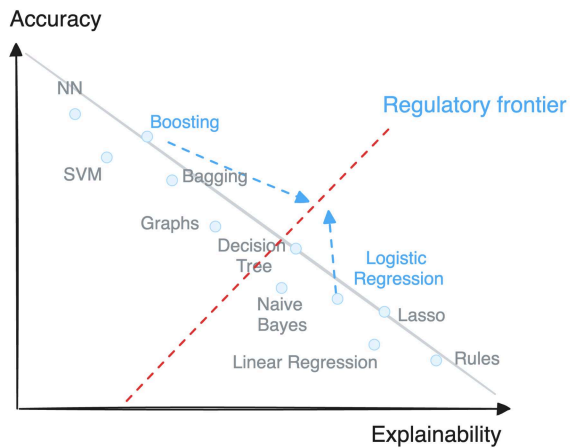
Monitoring and evaluation of these models is vital to maintain accountability and ensure that they continue to function as intended.

Adversarial debiasing ★

One technique we're using to vet AI models is called **adversarial debiasing**. It pits two machine learning models against each other: the first model predicts creditworthiness and the second one tries to predict the race, gender, or other potentially-protected class attributes of the applicant scored by the first model. Competition in this game drives both models to improve their methods until the predictor can no longer distinguish the race or gender outputs of the first model, resulting in a final model that is accurate and fair.

- [RAND Bayesian Improved Surname Geocoding](#)
- [Towards a Standard for Identifying and Managing Bias in Artificial Intelligence](#)
- [Artificial Intelligence Risk Management Framework \(AI RMF 1.0\)](#)
- [Investor POV: AI governance and responsible use | Insight Partners](#)
- [Explainability & Fairness in Machine Learning for Credit Underwriting](#)
- [Using AI to underwrite? You need to lock your AI](#)
- [Artificial intelligence: Experts propose guidelines for safe systems](#)
- [Credit Loan Decisions — Fairlearn 0.10.0.dev0 documentation](#)
- [Why Are We Using Black Box Models in AI When We Don't Need To? A Lesson From an Explainable AI Competition](#)
- [Fairness and machine learning](#)
- [National Institute of Justice Recidivism Forecasting Challenge Team "MCHawks" Performance Analysis · CrimRxiv](#)
- [Balancing False Positives | Andrew Wheeler](#)

Algorithms



Decision Trees

Gini impurity (L. Breiman et al. 1984) is a measure of non-homogeneity. It is widely used in classification trees.

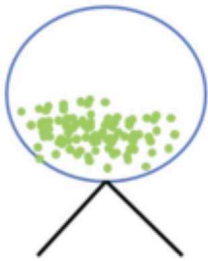
- [Decision Trees — Applied Machine Learning in Python](#)
- [31. Decision Trees in Python | Machine Learning](#)
- [11.2 Splitting Criteria | Practitioner's Guide to Data Science](#)

When the impurity increases, the purity decreases, hence also the entropy increases

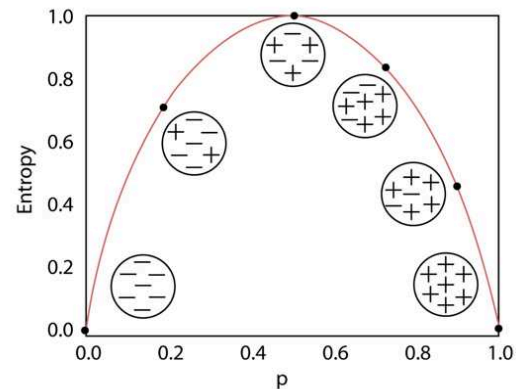
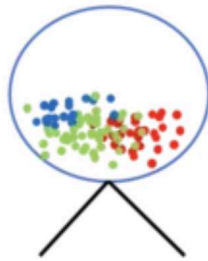
Gini Index works well in practice. Cross-Entropy is the entropy of the distribution over classes. You want to minimize the entropy over the classes, you want to have a very spike distribution in the classes, you want mostly to be one class and less of the others so that the trees are very certain about what the class is for each leaf.

Gini formula: $1 - \sum p^2 + (1-p)^2$

Totally pure



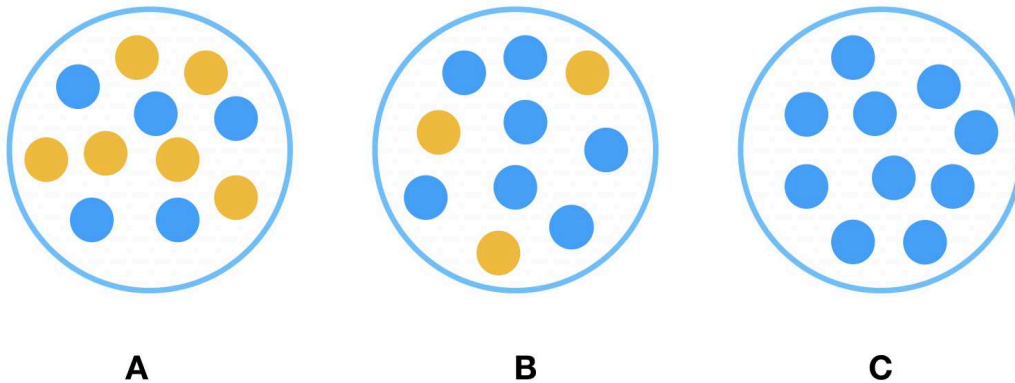
More impure



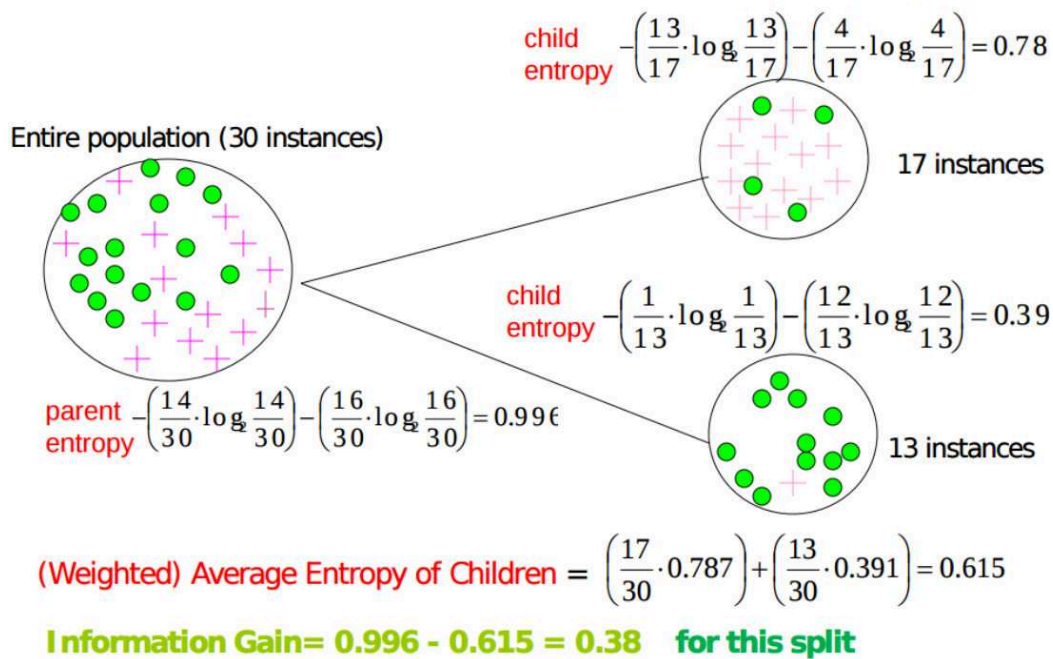
When a decision tree is defined with a target variable and the Best Split algorithm is applied, the algorithm aims to partition the data so that the resulting group of records at the new node minimizes impurity. A node with high impurity has a high population of several different values of the target variable because the parent split has not segmented the data effectively.

When you minimize impurity, you want the observations in each node to have the same value of the target variable. The homogeneity or purity of a partition increases with the proportion of observations that share the same target value. The Best Split algorithm in Xpress Insight uses the measure of [Gini impurity](#), which calculates the heterogeneity or impurity of the node. When the Gini impurity value is 0.0 (minimum value), the partition is homogeneous or pure. When the Gini impurity value is at its maximum value, the node is heterogeneous or impure. The maximum Gini impurity value varies for binary and multinomial target variables.

Looking at the samples in the following three nodes, which one is the easiest to describe? It is obviously C. Because all the samples in C are of the same type, so the description requires the least amount of information. On the contrary, B needs more information, and A needs the most information. In other words, C has the highest purity, B is the second, and A has the lowest purity. We need less information to describe nodes with higher purity.



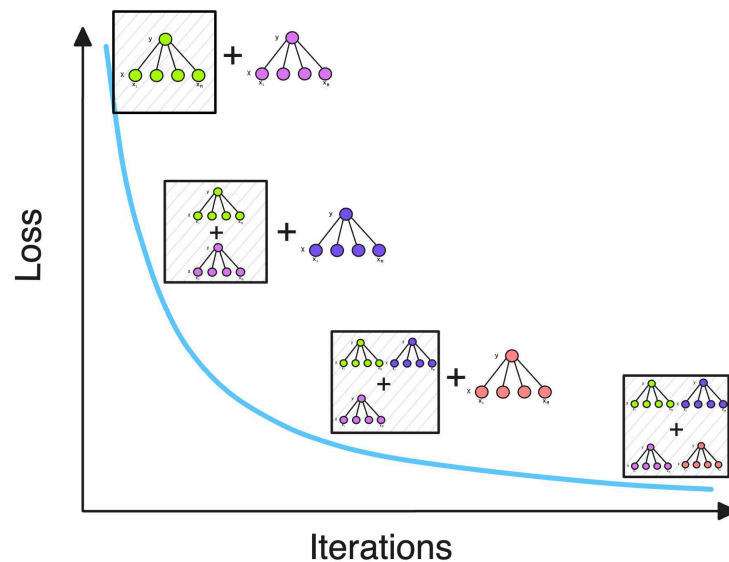
Information Gain = entropy(parent) – [average entropy(children)]



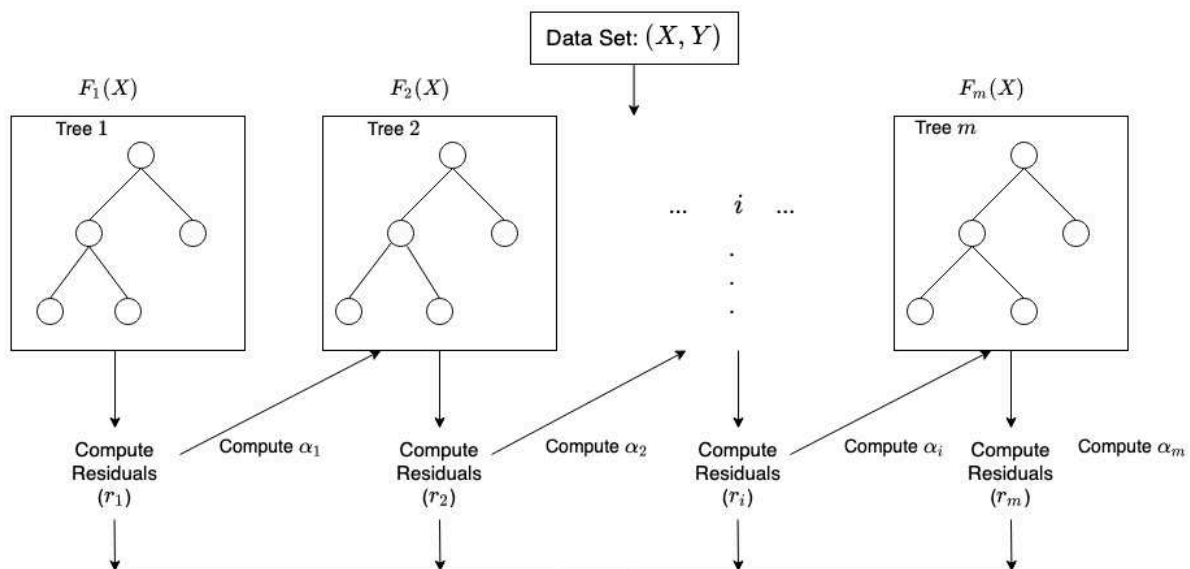
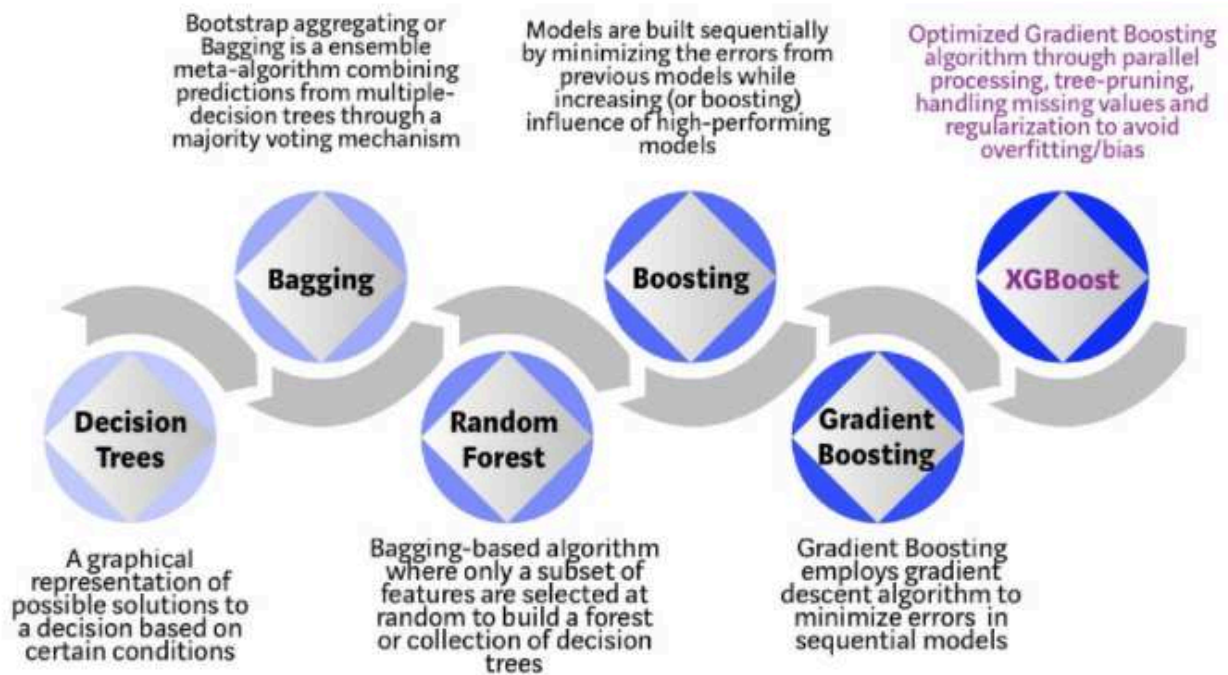
- [Why does frequency encoding work? - Data Science Stack Exchange](#)
- [Decision Trees Explained With a Practical Example – Towards AI](#)
- [Extending Scikit-Learn with CHAID model type - Openscoring](#)
- [A Lightweight Decision Tree Framework supporting regular algorithms: ID3, C4.5, CART, CHAID](#)
- [Machine Learning with Graphs | Course | Stanford Online](#)
- [Time Based Cross Validation](#)
- [Best Graph Neural Network architectures](#)

- [Graph Convolutional Networks \(GCN\)](#)
- [A Gentle Introduction to Graph Neural Networks](#)
- [Fast Track Introduction to Python for Machine Learning Engineers](#)
- [SAI #15: What's in Kubernetes for MLOps?](#)
- [Linear trees in LightGBM: how to use? | by Alessandro Morita Gagliardi | Medium](#)

Gradient Boosting



Note that, even for classification tasks, the trees that we build are regression trees! They are not classification trees. Indeed, we need the trees to predict the gradients, and the gradients are continuous.



$$F_m(X) = F_{m-1}(X) + \alpha_m h_m(X, r_{m-1}),$$

where α_i , and r_i are the regularization parameters and residuals computed with the i^{th} tree respectively, and h_i is a function that is trained to predict residuals, r_i using X for the i^{th} tree. To compute α_i we use the residuals computed, r_i and compute the following: $\arg \min_{\alpha} = \sum_{i=1}^m L(Y_i, F_{i-1}(X_i) + \alpha h_i(X_i, r_{i-1}))$ where $L(Y, F(X))$ is a differentiable loss function.

Gradient Boosting Model

M0

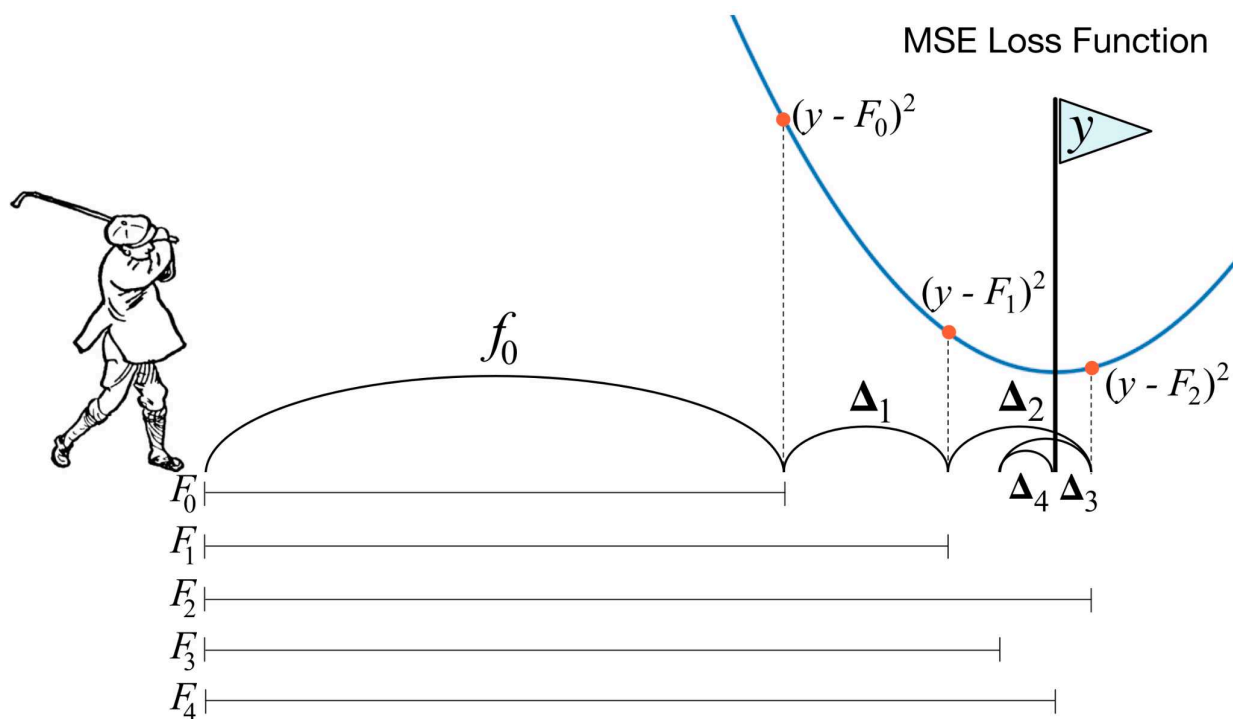
M1

M2

...

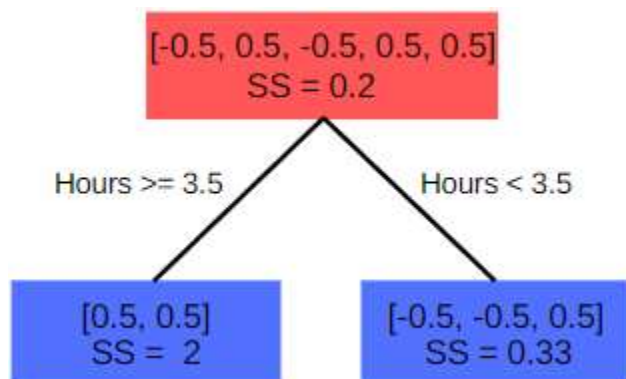
Mn

$$\begin{aligned}
 F_0(X) &= \gamma_0 H_0(x, y) + e_0 \\
 F_1(X) &= F_0(X) + \gamma_1 H_1(x, e_0) + e_1 \\
 F_2(X) &= F_1(X) + \gamma_2 H_2(x, e_1) + e_2 \\
 &\vdots \\
 F_n(X) &= F_{n-1}(X) + \gamma_n H_n(x, e_{n-1}) + e_n
 \end{aligned}$$



- XGBoost
- Regression Loss - XGBoost

- [Understanding Gradient Boosting as a gradient descent | Nicolas Hug](#)
- [Gradient boosting \(Notebook\)](#)
- [Introduction to Boosted Trees](#)
- [Notes on Logistic Regression and Logit Boost](#)
- [Additive Logistic Regression: a Statistical View of Boosting](#)
- [Gradient Boosting \(Continued\)](#)
- [XGBoost | Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining](#)
- [Tree Boosting With XGBoost](#)
- [Explanation of min_child_weight in xgboost algorithm - Cross Validated](#)
- [Gradient Boosting for Linear Regression - why does it not work? - Cross Validated](#)
- [Is a decision stump a linear model? - Cross Validated](#)
- [Gradient boosting: frequently asked questions](#)
- [Gradient boosting performs gradient descent](#)
- [What Is Gradient Boosting In Machine Learning | CitizenSide](#)
- [XGBoost | Machine Learning - Pianalytix: Build Real-World Tech Projects](#)
- [Understanding Gradient Boosting Tree for Binary Classification](#)



$$f_1(x) \approx y$$

$$f_2(x) \approx y - f_1(x)$$

$$f_3(x) \approx y - f_1(x) - f_2(x)$$

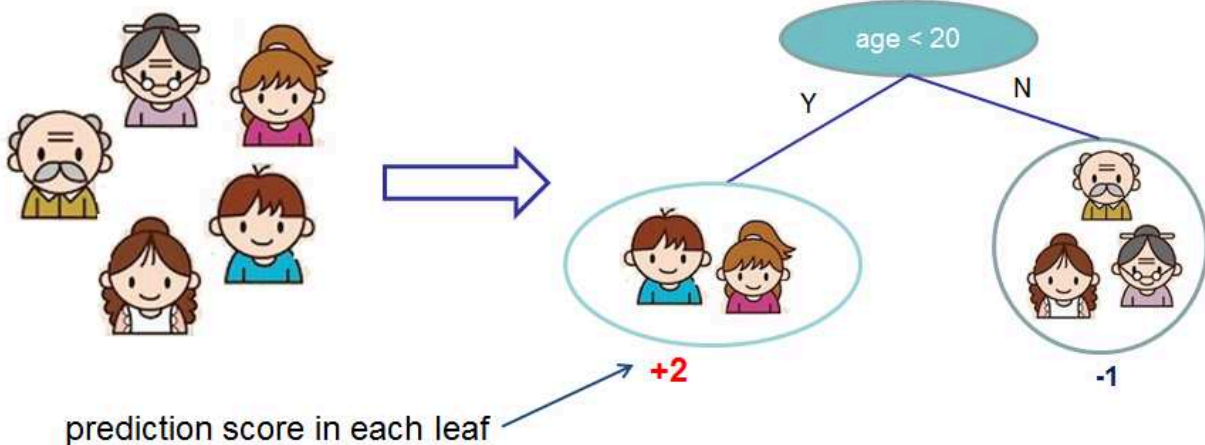
$$\mathcal{L}_{split} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma$$

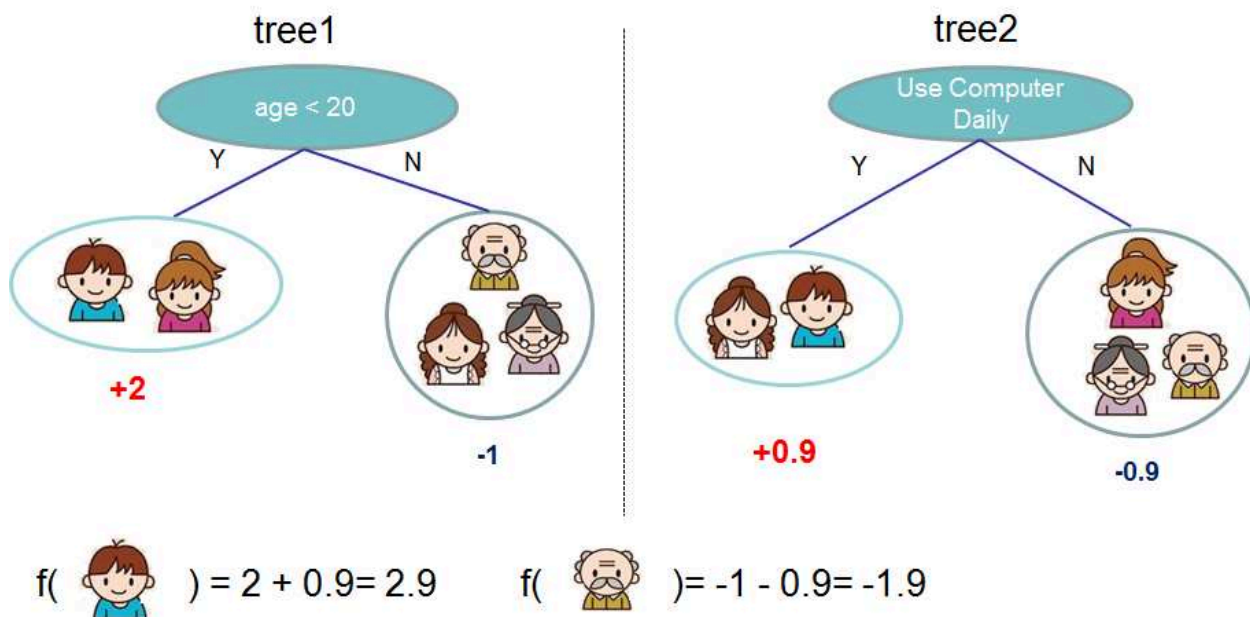
If you have a tree structure where each branch consists of a root node, an internal node, and two leaf nodes like $x_1 < 5$ and $x_2 < 2$, and if these splits are the primary interactions considered in the tree, then you would have a total of 4 such interactions.

Each branch represents one combination of feature interactions, and with 2 leaf nodes per branch, you have 4 different combinations in total. So, in this specific tree structure, you would have 4 interactions of this kind.

Input: age, gender, occupation, ...

Like the computer game X





Gradient Boosted Trees

- [Understanding XGBoost | Towards Data Science](#)
- [Introduction to Boosted Trees — xgboost 2.1.0-dev documentation](#)
- [Demystify Modern Gradient Boosting Trees](#)
- [Appendix to Gradient Boosting and XGBoost](#)
- [How Does Extreme Gradient Boosting \(XGBoost\) Work?](#)
- [How to compute the gradient and hessian of logarithmic loss ...](#)
- [How does scale_pos_weight affect probabilities? - XGBoost](#)
- [XGBoost Best Iteration - Stack Overflow](#)
- [how to plot XGBoost evaluation metrics? - Stack Overflow](#)
- [Understanding Gradient Boosting Tree for Binary Classification](#)
- [Information Gain](#)
- [How to make LightGBM to suppress output? - Data Science Stack Exchange](#)
- [Cost-sensitive boosted tree for loan evaluation in peer-to-peer lending - ScienceDirect](#)
- [imbalance-xgboost · PyPI](#)
- [Construction of a financial default risk prediction model based on the LightGBM algorithm](#)
- [Understanding XGBoost | Towards Data Science](#)

- [XGBoost colsample_by* hyperparameters explained | by MLComrade | Analytics Vidhya | Medium](#)
- [Demo for prediction using individual trees and model slices — xgboost 2.1.0-dev documentation](#)
- [xgboost: “Hi I’m Gamma. What can I do for you?” — and the tuning of regularization | by Laurae | Data Science & Design | Medium](#)

Tree Parsing

- [Parsing](#)
- [How to get each individual tree's prediction in xgboost? - Stack Overflow](#)
- [What is an intuitive interpretation of the leaf values in XGBoost](#)
- [LightGBM to SQL](#)
- [XGB to SQL](#)
- [XGBoost leaf scores - python](#)

Boosted Scorecards

- [XGBoost for Interpretable Credit Models | credit_scorecard – Weights & Biases](#)
- [Credit_Scorecards_with_XGBoost_and_W&B.ipynb \[Notebook\]](#)
- [WANDB Artifacts - Credit Scorecard \[Utils files\]](#)
- [W&B credit_scorecard project \[Repository for the project\]](#)
- [showcase/event_notebooks/GTC_2021/credit_scorecard](#)
- [Be Careful When Interpreting Predictive Models in Search of Causal Insights | by Scott Lundberg | Towards Data Science](#)
- [Evaluating Risk for Loan Approvals using XGBoost \(0.90\)](#)
- [File:Roc-draft-xkcd-style.svg - Wikimedia Commons](#)

Random Forest

- [Random Forests\(TM\) in XGBoost](#)
- [Beware Default Random Forest Importances](#)
- [rf boosting type for classification #4118 - microsoft/LightGBM - GitHub](#)
- [A two-stage credit scoring model based on random forest: Evidence from Chinese small firms - ScienceDirect](#)

- [Random Forest in Python](#)
- [Credit scorecard model - Data Science Stack Exchange](#)
- [Can't see the forest for the trees | SpringerLink](#)
- [Application Scorecard using Deep Learning](#)
- [Guide to Credit Scoring in R](#)
- [XGBoost Alternative Base Learners | by Corey Wade | Towards Data Science](#)
- [scikit-learn/sklearn/ensemble/_forest.py](#)
- [RF splitter in sklearn](#)

In scikit-learn, at each node of a tree, we will visit at random N first feature where $N = \text{int}(X.\text{shape}[0] * \text{max_features})$. It corresponds to the following code: [github.com](#)

```
Python
# Sample up to max_features without replacement using a
# Fisher-Yates-based algorithm (using the local variables `f_i` and
# `f_j` to compute a permutation of the `features` array).
#
# Skip the CPU intensive evaluation of the impurity criterion for
# features that were already detected as constant (hence not suitable
# for good splitting) by ancestor nodes and save the information on
# newly discovered constant features to spare computation on descendant
# nodes.
while (f_i > n_total_constants and # Stop early if remaining features
                                   # are constant
        (n_visited_features < max_features or
         # At least one drawn features must be non constant
         n_visited_features <= n_found_constants + n_drawn_constants)):
    n_visited_features += 1
```

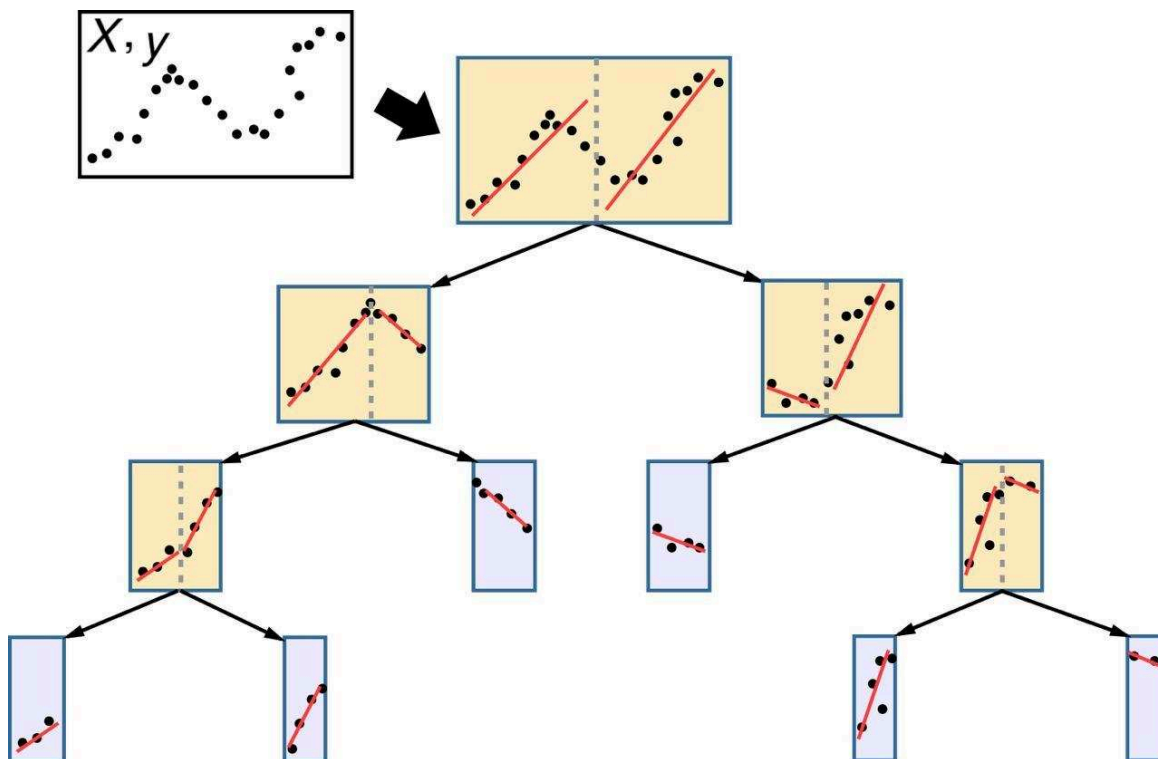
The Fisher-Yates algorithm ([Fisher-Yates shuffle - Wikipedia](#)) is a way to explore randomly the features for the quantity provided by `max_features`.

Bias Variance Trade-Off

- [The “Bias-Variance Trade-Off” Explained Practically \(In Python\) | by Samuele Mazzanti | Towards Data Science](#)
- [Stop Using The Same Image in Bias-Variance Trade-off Explanation | by Ivan Reznikov | Medium](#)

- [bias_variance_decomp](#): Bias-variance decomposition for classification and regression losses - mlxtend
- [mlxtend/mlxtend/evaluate/bias_variance_decomp.py at master · rasbt/mlxtend · GitHub](#)

Linear base learners



[bias_variance_decomp](#): Bias-variance decomposition for classification and regression losses - mlxtend

“The Quest for the Grail of Machine Learning”, the overfitting and underfitting problem, sometimes referred to as the bias-variance trade-off, which is the balancing between the model’s complexity and the amount of input data. The presence of either overfitting or underfitting are indicators of a possible imbalance between the model’s complexity and the amount of input data.

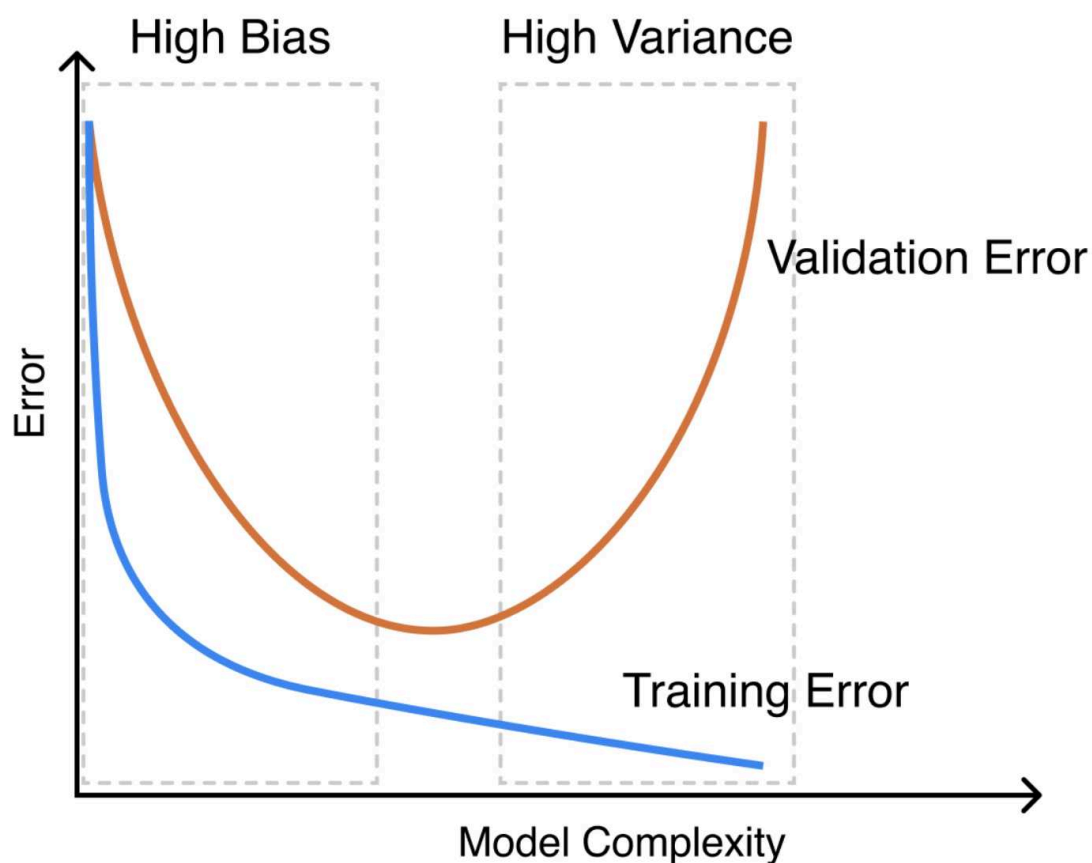
When the model is biased, we call it underfitting.

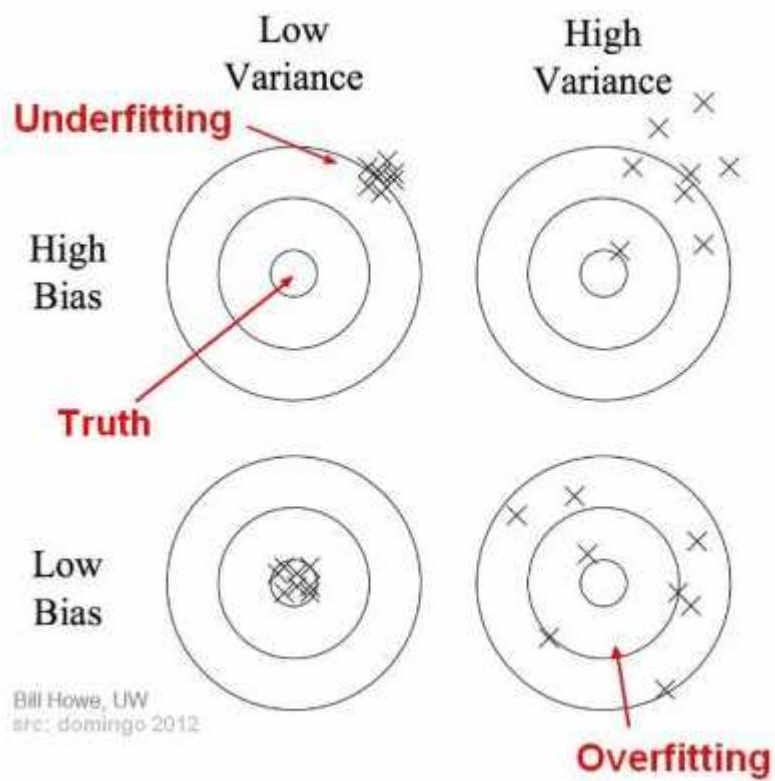
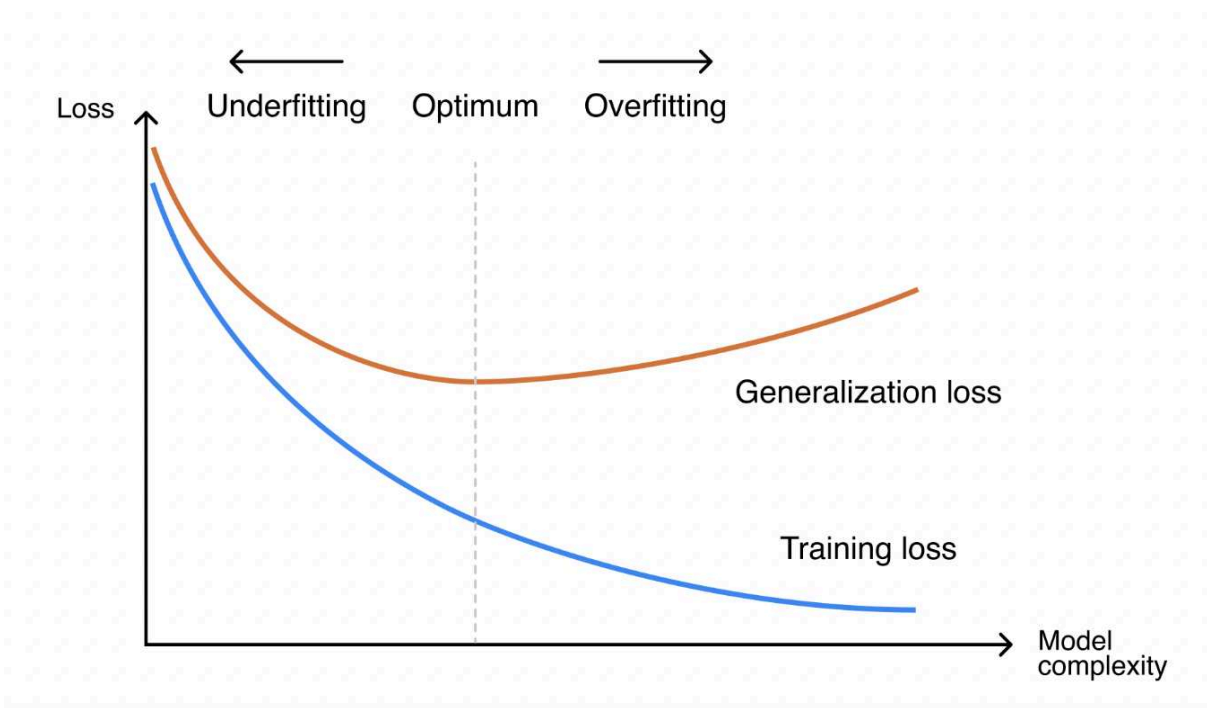
That is the essence of machine learning: we train a model using specific loss, we measure its performance using different sets of metrics in the hope of achieving something different from the first and second, and sooner or later we will face a plateau, where the first or even second element improves, but the third does not react.

We must be familiar with business cases and the nature of data to evaluate all possible pitfalls. It is a good practice to collect 10–20 objects with the largest residuals after training as attendant artifacts.

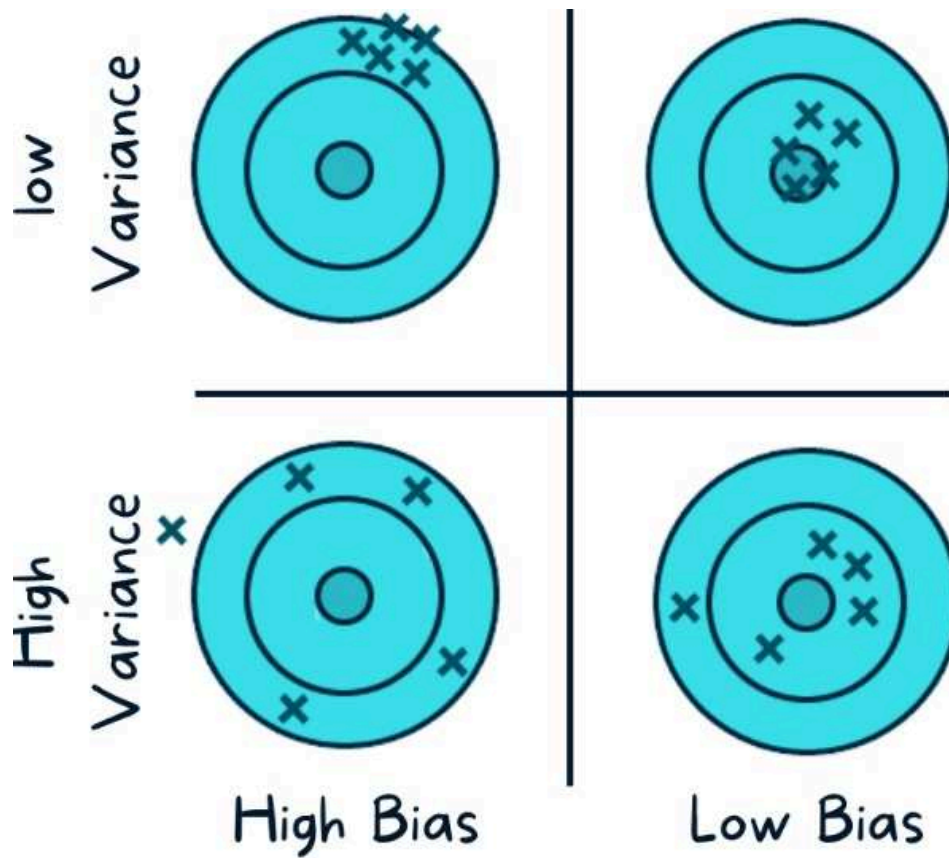
Adversarial learning: a “machine learning model for analyzing machine learning models”. Worst case analysis: we set 0 for “good” samples of data and 1 for “bad” samples (e.g., taking the top-N% biggest residuals in the second case). We should try different thresholds for our particular set. We fit a simple classifier (e.g., logistic regression) on these labels and calculate ROC-AUC. If two classes are separable (AUC is significantly greater than 0.5), then we analyze the model’s weights, and this gives us a hint of which exact features distinguish our “worst” cases from others the most.

The corner-case analysis aims to test the model in rare circumstances.

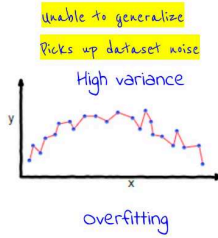




$\text{error} = \text{bias} + \text{variance}$



Bias-Variance Tradeoff

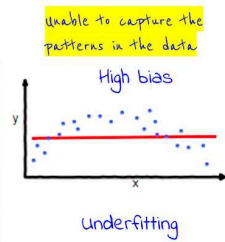


Symptoms:

- Training error much lower than test error
- Training error lower than the expected error

Remedies:

- Add more training data
- Use simpler models
- Use bagging

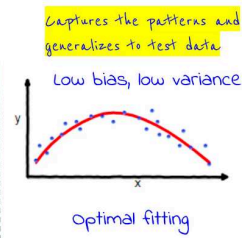


Symptoms:

- Training error is higher than expected (often testing error too)

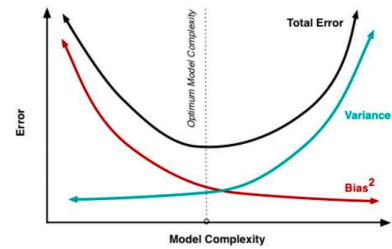
Remedies:

- Use more complex model (e.g. linear to non-linear)
- Use more features
- Use boosting



Properties:

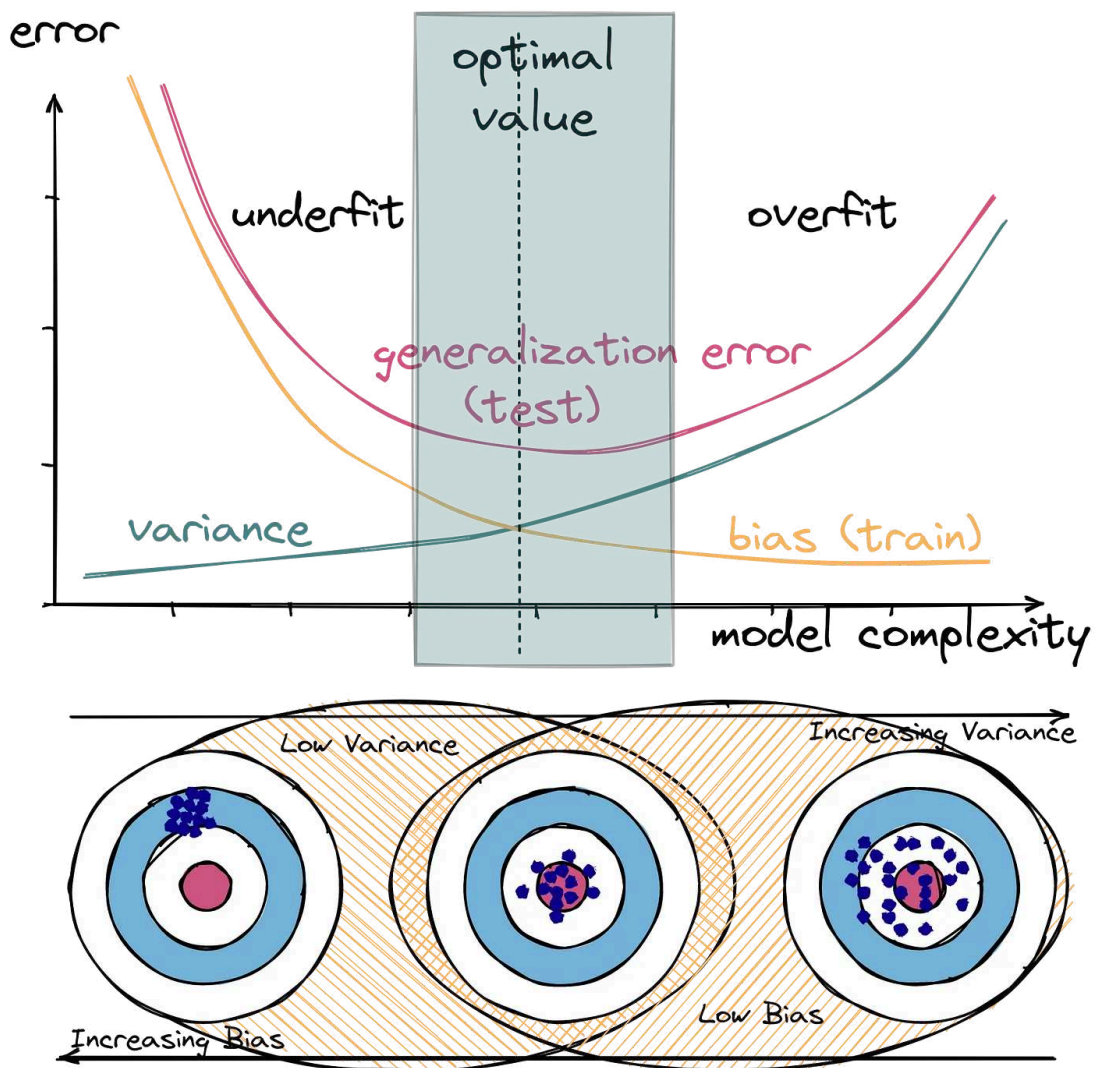
- Training and testing errors are similar and closer to the expected error



$$\text{Tot. Err} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

- when you try to reduce the bias too much, it increases the variance
- when you try to reduce the variance too much, it increases the bias
- need to find the right balance to reduce the overall error
- There will always be some error due to the noise in the dataset

References: <http://scott.fortmann-roe.com/docs/BiasVariance.html> | <https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>

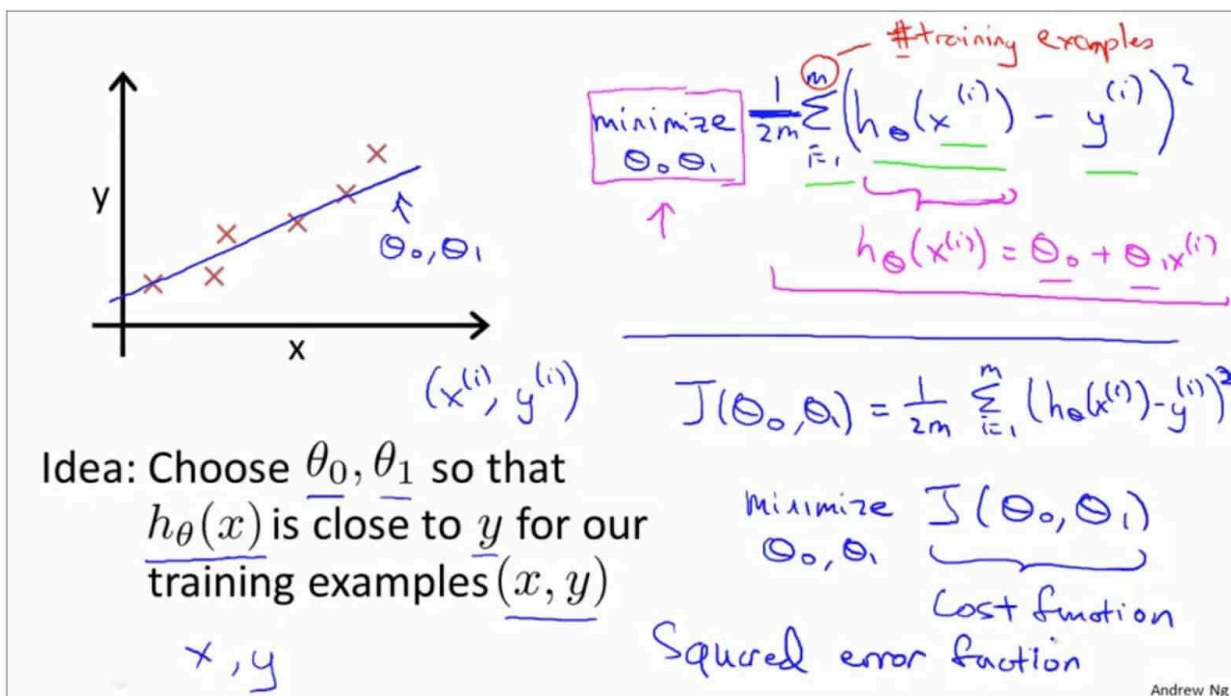


Boosting is based on weak learners (high bias, low variance). In terms of decision trees, weak learners are shallow trees, sometimes even as small as decision stumps (trees with two leaves). Boosting reduces error mainly by reducing bias (and also to some extent variance, by aggregating the output from many models).

On the other hand, Random Forest uses fully grown decision trees (low bias, high variance). It tackles the error reduction task in the opposite way: by reducing variance. The trees are made uncorrelated to maximize the decrease in variance, but the algorithm cannot reduce bias (which is slightly higher than the bias of an individual tree in the forest). Hence the need for large, unpruned trees, so that the bias is initially as low as possible.

Loss Functions

Cost Function



Thus, to align residuals to a specific loss, one may prefer using a single term of the loss sum as a pseudo-residuals instead of raw differences, which are a squared error for MSE, a logarithmic error for RMSLE, or a class label multiplied by the predicted probability of this class for LogLoss.

Working with highly imbalanced data — Applied Machine Learning in Python

- [Focal loss implementation for LightGBM](#) • Max Halford
- [Focal_loss_on_iFocal loss implementation for LightGBM](#) • Max Halfordnit_score.ipynb
- [LightGBM/advanced_example.py](#)
- [LightGBM with the Focal Loss for imbalanced datasets](#) | by Javier Rodriguez Zaurin | Towards Data Science
- [LightGBM Custom Loss Function - Python](#)
- [An implementation of the focal loss to be used with LightGBM for binary and multi-class classification problems](#)
- [Calibrating Deep Neural Networks using Focal Loss](#)
- [Practical Lessons from Predicting Clicks on Ads at Facebook](#)

- [Calibration of probabilities for tree-based models | Guilherme's Blog](#)
- [Facebook Practical Lessons](#)
- [Practical Lessons from Predicting Clicks on Ads at Facebook · Papers I Read](#)
- [R/utils.R](#)
- [Recalibration after Negative Sampling | Johannes Haupt](#)
- [Metric Names: Relative Information Gain vs. LogLoss Reduction · Issue #2689 · dotnet/machinelearning · GitHub](#)
- [Making Sense of Logarithmic Loss](#)
- [Log Loss Function Explained by Experts](#)
- [Binary cross-entropy and logistic regression | by Jean-Christophe B. Loiseau | Towards Data Science](#)
- [Decoding Logistic Regression Using MLE - Analytics Vidhya](#)
- [Maximum Likelihood the Easy Way. A complex topic simplified | by Brian Pietracatella | Towards Data Science](#)
- [11. Logistic Regression — Advanced Topics in Computational Physics](#)
- [Logistic Regression - Statistics & Data Science](#)
- <https://github.com/boutrosrg/Regression-Analysis/tree/master/Logistic%20Regression>
- [Implementing logistic regression from scratch in Python](#)
- [FocalLoss_Ex.ipynb - suvo0000/Machine_Learning · GitHub](#)
- [GitHub - handongfeng/Xgboost-With-Imbalance-And-Focal-Loss](#)
- [jhwjhw0123/Imbalance-XGBoost: XGBoost for label-imbalanced data](#)
- [Transactional Network Analysis and Money Laundering Behavior Identification of Central Bank Digital Currency of China](#)
- [Logistic Regression - Data Automaton](#)
- [Partial derivative of MSE cost function in Linear Regression?](#)
- [XGBoost tree outputs](#)
- [Brier Score Decomposition](#)
- [Why binary crossentropy can be used as the loss function in autoencoders? - Cross Validated](#)
- [Chapter 1: Optimization & Gradient Descent - Tomas Beuzen](#)
- [Appendix B: Logistic Loss — Deep Learning with PyTorch](#)
- [Batch gradient descent versus stochastic gradient descent](#)
- [Cost function of neural network is non-convex? - Cross Validated](#)
- [22.7. Maximum Likelihood - Dive into Deep Learning](#)
- [AUC VS LOG LOSS — Data Machines](#)

A/B Testing

What is an A/B test? A tool that helps to make reliable decisions made on data. Also called Multivariate testing / Hypothesis testing / Experimentation.

Running too many tests dilutes the results.

Opportunity sizing → Drivers of change → User research → Learn from previous efforts

Hypothesis is something to be disproved. The null hypothesis, if you introduce a change → nothing happens.

Test duration & group size

Novelty vs Primacy Effects

- [How A/B Testing Works with Machine Learning | Capital One](#)
- [A/B Testing – Big Change, Not Big Bang | Capital One](#)

